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15 November 1982
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MIL-C-85437(AS)
25 November 1981

MILITARY SPECIFICATION
CAMERA SET, TELEVISION, AN/AXX-1

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes requirements for the manufacture and acceptance of Television Camera Set (TCS) AN/AXX-1 prime item.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-5088G 30 Sep 77	Wiring; Aircraft, Installation of
MIL-E-5400R 31 Oct 75	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422F(AS) 30 Nov 77	Testing, Environmental, Aircraft Electronic Equipment
MIL-M-7793D Supplement 1 31 Dec 69	Meter, Time Totalizing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Dept. (ESSD), Code 93, Lakehurst, NJ 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5821

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MIL-E-17555G Amendment 2 15 Apr 70	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303B 1 Sep 66	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307F 4 Aug 78	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-S-19500F Amendment 2 9 Jun 80	Semiconductor Device, General Specification for
MIL-M-38510D Supplement 1H 13 Feb 81	Microcircuits, General Specification for
MIL-I-46058C Amendment 5 3 Apr 79	Insulating Compound, Electrical (for Coating Printed Circuit Assemblies)
MIL-C-83723 27 Dec 77	Connector, Electrical (Circular, Environment Resisting), Receptacles, and Plugs

STANDARDS

Military

MIL-STD-461A Change 6 3 Jul 73	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-470 21 Mar 66	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-704B 1 Aug 75	Electric Power, Aircraft, Characteristics, and Utilization of
MIL-STD-781C 21 Oct 77	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-785B 15 Sep 80	Reliability Program for Systems and Equipment, Development and Production

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MIL-STD-794D Notice 4 8 Aug 75	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-883B Notice 4 5 Dec 80	Test Methods and Procedures for Microelectronics

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

AFSC DESIGN HANDBOOK

DH 1-4 5 Jan 76	Electromagnetic Compatibility, 3rd Edition, Revisions 1 and 2

NAVAL RESEARCH LABORATORY
SHOCK AND VIBRATION INFORMATION CENTER

Shock and Vibration Bulletin No. 41

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-170 - Electrical Performance Standard for Monochrome Television Studio Facilities.

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, N.W., Washington, DC 20006.)

NORTHROP CORPORATION

WS-1308 - Norcote, Moisture- and Fungus-Resistant Coating.
MP-101 - Fabrication and Inspection of Printed Wiring Boards.

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DRAWINGS

336241	-	Reliability Test Plan for TVSU Program.
336256	-	Reliability Test Procedure, Reliability Qualification Test for Television Camera Set (TCS).
336700	-	Acceptance Test Procedure for Television Camera Set.
339000	-	Camera Set, Television.
339005	-	Wiring Diagram Interface Drawing.
339009	-	Television Camera Assembly Installation Control Drawing.
339010	-	Control-Power Supply Installation Control Drawing.
339050	-	Television Camera Assembly.
339500	-	Control-Power Supply Assembly.
339735	-	Module Assembly - Video Output

(Application for copies should be addressed to Northrop Corporation, Electro-Mechanical Division, 500 East Orangethorpe Avenue, Anaheim, California 92801.)

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3. REQUIREMENTS

3.1 System description. The TCS shall provide the F-14 aircraft with a visual target identification system that offers a 10:1 nominal improvement in resolution within the positive identification range of the unaided eye of the aircrew during clear weather and daylight conditions.

3.1.1 Major components list. The TCS major components are as follows: (see Drawing 339000)

Weapon-Replaceable Assemblies (WRA)	Measurement	Max. Weight (lb)
WRA-1 Television Camera Assembly (TCA) Drawing No. 339050	Length shall be not greater than 31.8 inches. Diameter shall be not greater than 9.0 inches. (see Drawing 339009)	60
WRA-2 Control-Power Supply (CPS) Drawing No. 339500	Length shall be not greater than 16.0 inches. Height and width shall be not greater than 8.0 inches each. (see Drawing 339010)	40
	Total	100

3.2 First article. When specified, a sample shall be subjected to first article inspection. (See 4.4.2.4 and 6.6.)

3.3 Characteristics

3.3.1 Detail performance requirement

3.3.1.1 Television Camera Assembly (TCA). The TCA shall meet the requirements listed in the following paragraphs.

3.3.1.1.1 Function. The TCA shall convert target images into television images at two selectable magnifications while providing two axis rate stabilization of the television cameras lines of sight.

3.3.1.1.2 Form factor. The TCA shall be mounted and enclosed in a cylindrical case which will attach to the F-14 airframe. The enclosure shall be not greater than 31.8 inches in length nor 9.0 inches diameter.

3.3.1.1.3 Weight. The weight of the TCA shall be not greater than 60 pounds.

3.3.1.1.4 Contents. The TCA shall contain the following components: vidicons, telescopes, gimbals, and the associated electronics of the TCA.

3.3.1.1.5 Controls. All controls for the TCS are located in the F-14. These controls are not provided as part of the TCS system. Control signals

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will be provided to the TCS interface. The TCS shall, when specified power is provided, respond to all proper control signals arriving at the TCS-F-14 interface, shown in Drawing 339005. Control signals are defined in 3.3.2.17.

3.3.1.1.6 Electrical connections. External electrical connectors on the TCS shall meet the requirements of MIL-E-5400 or MIL-C-83723 and shall also be as called out on Drawing 339005.

3.3.1.1.7 Warmup time. The time required after initial turn-on for the TCS to perform as specified shall not be greater than two minutes from a 25°C ambient condition and not greater than 9 minutes from a -18°C ambient condition.

3.3.1.2 Control-Power Supply (CPS). The CPS shall contain regulated dc power supplies and electronics required for synchronization signal generation, video tracker logic, video symbology generation, mode control, position control, and interface buffers.

3.3.1.2.1 Form factor. The CPS shall be mounted internal to the F-14 and shall be 16.0 inches long by 8 inches wide by 8 inches high.

3.3.1.2.2 Weight. The weight of the CPS shall not be greater than 40 pounds.

3.3.1.2.3 CPS performance. The CPS shall include the interface between: the TCS and the F-14 air vehicle; the AN/AWG-9 radar; and the AN/AWG-9 computer. The source, destination, and characteristics of these interface signals, their purpose, and interaction with other onboard systems and components are specified in 3.3.2.

3.3.1.3 Television Camera Set (TCS). The TCS shall consist of the TCA and the CPS. The following paragraphs specify the modes of operation of the TCS as installed in the F-14 aircraft.

3.3.1.3.1 Tracking mode. The TCS shall be capable of automatically video tracking any airborne target within its cone of vision which has an apparent contrast, at the system input, of not less than 20 percent. Contrast is defined as the brightness of the target (B_T) minus the brightness of the background (B_B) divided by the brightness of the background:

$$\frac{B_T - B_B}{B_B}$$

3.3.1.3.2 Acquisition modes. The TCS shall have four selectable modes for target acquisition. These modes shall be: manual acquisition; automatic acquisition; automatic acquisition with search; and semiautomatic acquisition.

3.3.1.3.2.1 Manual acquisition. When the manual acquisition mode has been selected, the TCS shall provide the operator with the capability to manually acquire and lock on to a target with an aspect ratio of approximately 1:1 and exhibiting a stable image not less than two TV lines in height. The manual acquisition mode shall be capable of being entered at any time from any mode. Entry into the manual acquisition mode shall be made by first entering the manual search capability. Entry into the manual search capability shall be accomplished by positioning the switch on the hand control to the first detent (half action) which shall cause the system to respond to rate commands gener-

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ated by the hand control unit, thereby enabling manual search capability. Once the operator has selected a target and positioned the track window over the selected target, manual acquisition shall be entered by positioning the switch on the hand control to the second (full action) detent. Acquisition time during the manual mode is a function of the availability of targets and the skill of the individual operator. Appendix A is a chart of the mode control switches and their function.

3.3.1.3.2.2 Automatic acquisition. When the automatic acquisition mode has been selected, the TCS shall automatically acquire a stable target having an aspect ratio of approximately 1:1 and within the center 50 percent of the raster. The target must exhibit an image at least three TV lines high in an uncluttered uniform background. Entry into the automatic acquisition mode shall be made by selecting the TCS slaved mode/condition and the automatic acquisition mode. During this mode, the TCS shall be slaved to either the AWG-9 radar or the computer. Acquisition time in this mode is 0.75 second maximum, exclusive of slew time. The automatic acquisition, automatic search, and stick mode flow diagrams, shown in Figures 1, 2, and 3, illustrate the sequence of events required to achieve lock-on and track.

3.3.1.3.2.3 Automatic acquisition with search. When the automatic acquisition with search mode has been selected, the TCS shall enter a mode that augments the automatic acquisition mode. In this mode, a search pattern, generated internally by the TCS, will be added to the externally-applied slaving commands to effectively increase the acquisition field of view. During the automatic acquisition with search mode, the acquisition field of view shall be effectively increased in area by a minimum factor of 6.0 when the 3 x 3 search pattern is used, and by a minimum factor of 15.0 when the 5 x 5 search pattern is used.

3.3.1.3.2.4 Search patterns. Two search patterns shall be generated by the TCS, as shown in Figures 4 and 5. These patterns shall be generated when the automatic-search mode is selected and the size of the pattern generated shall be a function of the Search Pattern Size (SPATS) signal specified in 3.3.2.12. Step sizes between positions in the search patterns presented in Figures 4 and 5 are 4.8 mrad (16.5 minutes) for the narrow field of view, and 15.8 mrad (54.3 minutes) for the wide field of view $\pm$11%. The time required for search and acquisition shall be a time delay of 0.75 second or less, plus 267 msec for each step required to locate the target.

3.3.1.3.2.5 Semiautomatic acquisition. The semiautomatic mode shall be entered from either the automatic-acquisition or autosearch mode. The TCS shall acquire a stable target having an aspect ratio of approximately 1:1 and within the center 50 percent of the raster. The target must exhibit an image at least three TV lines high in an uncluttered uniform background. This mode shall be initiated by the operator in the same manner as manual acquisition except that the target need only be placed within the inner 50 percent of the selected field of view. Acquisition time for semiautomatic acquisition shall be identical to the manual-acquisition mode.

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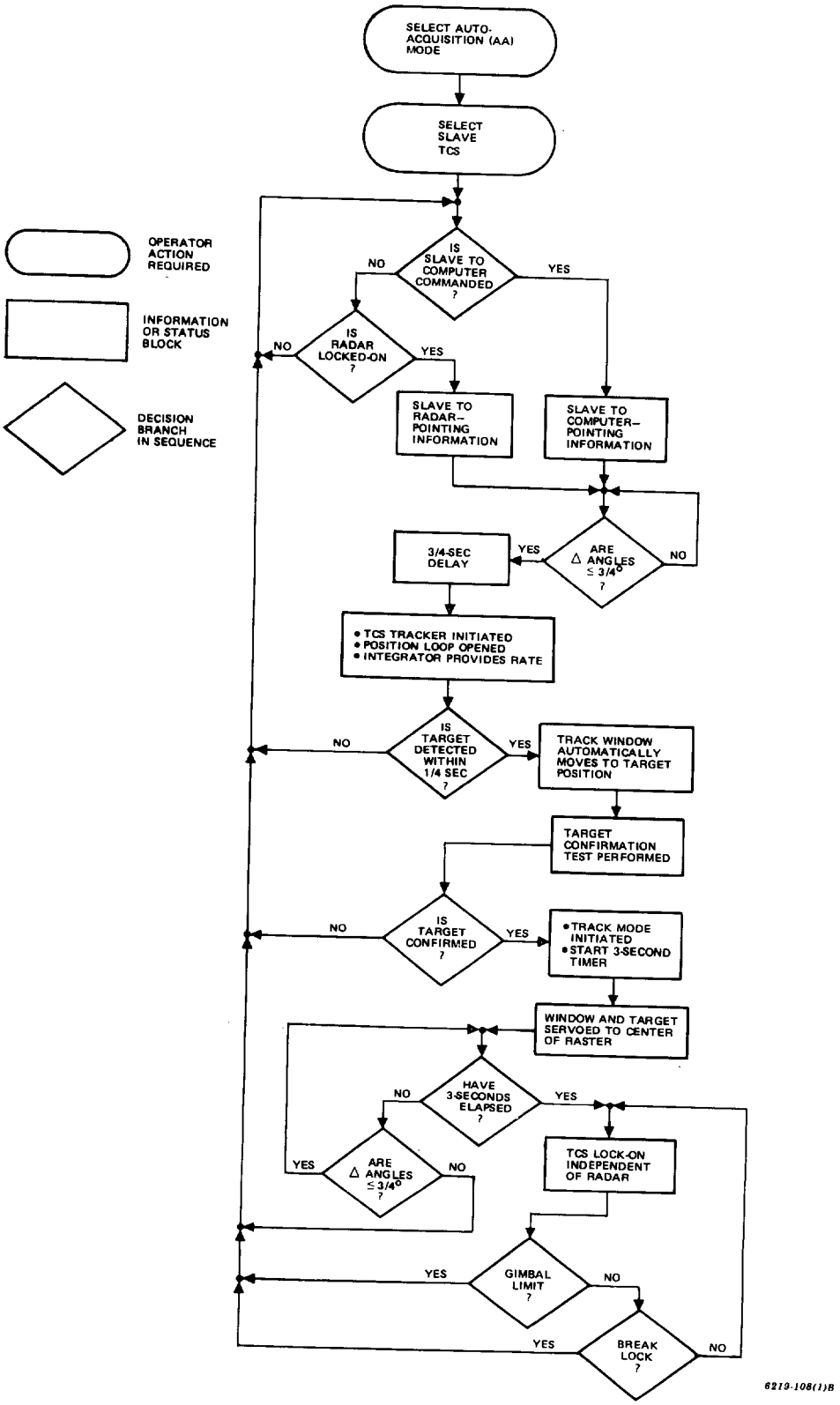


FIGURE 1. TCS flow diagram - Auto-Acquisition (AA) mode.

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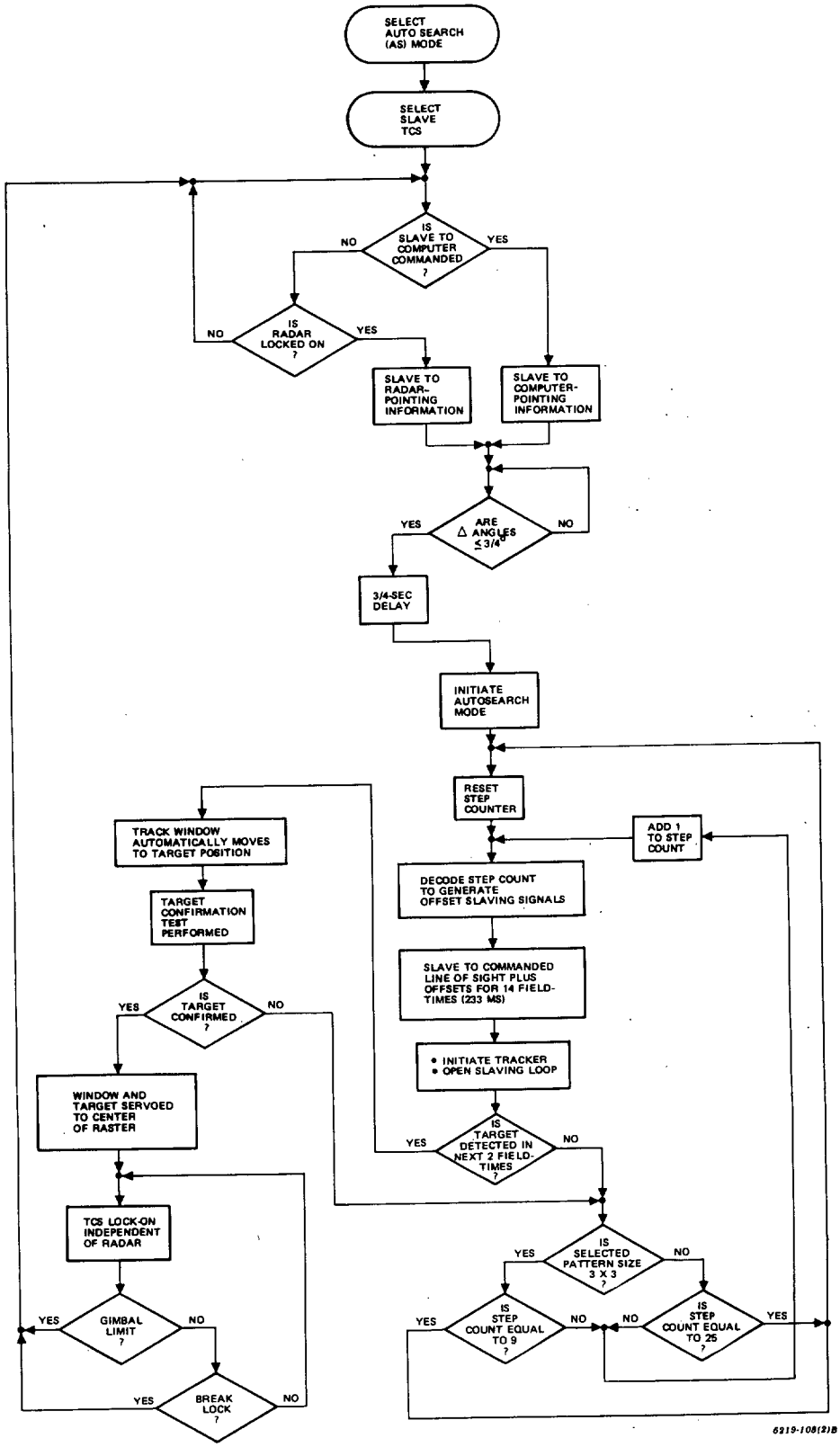


FIGURE 2. TCS flow diagram - Auto-Search (AS) mode.

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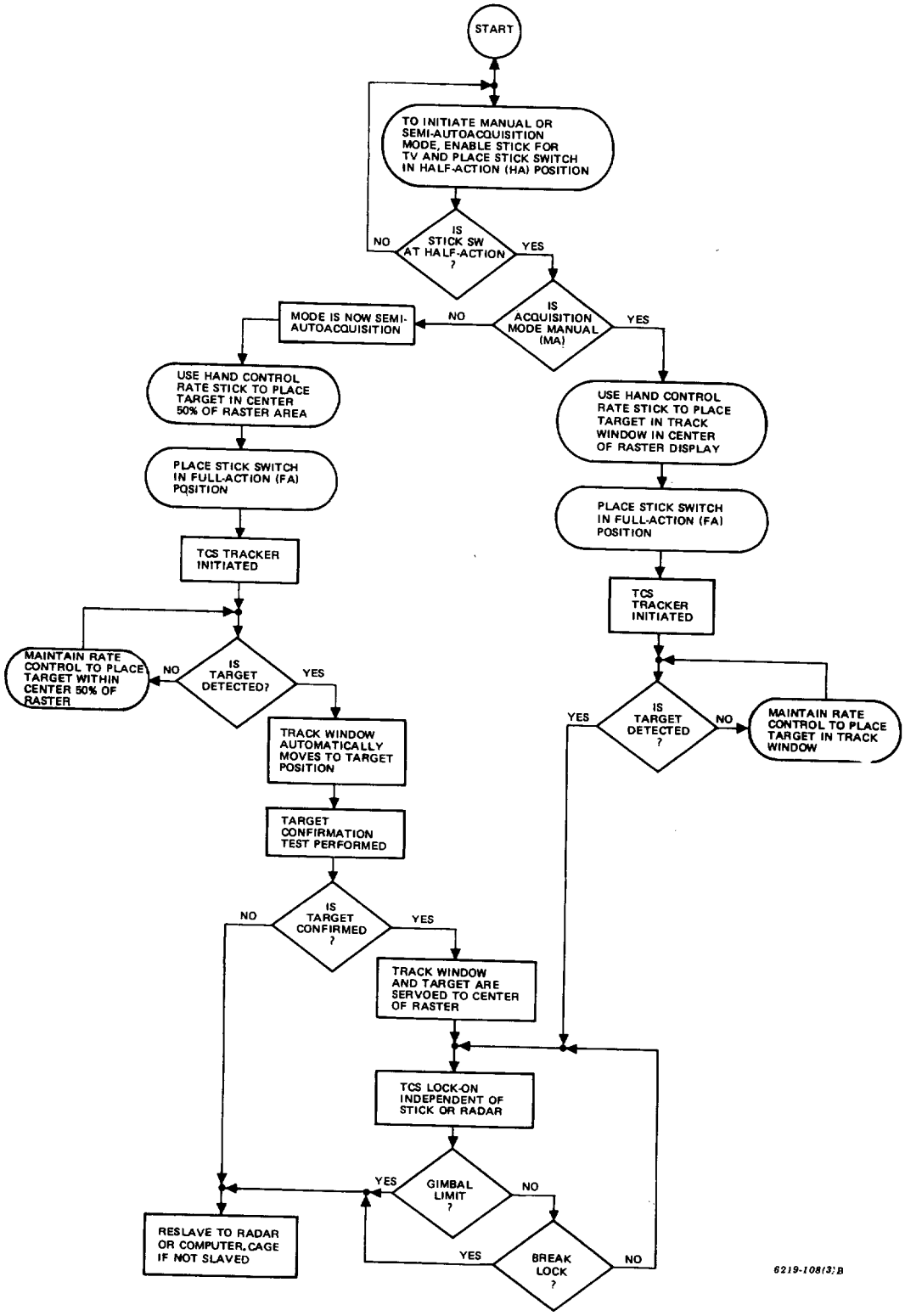


FIGURE 3. TCS flow diagram - stick modes.

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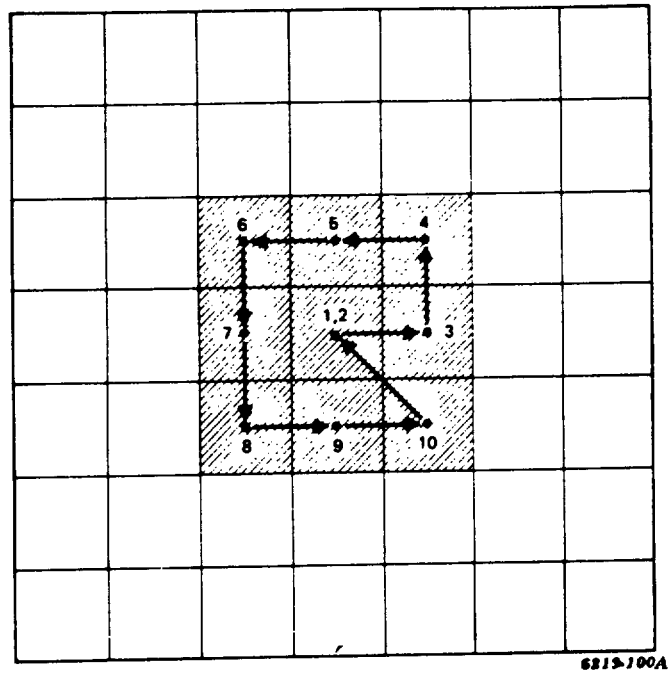


FIGURE 4. Scan pattern 3 x 3.

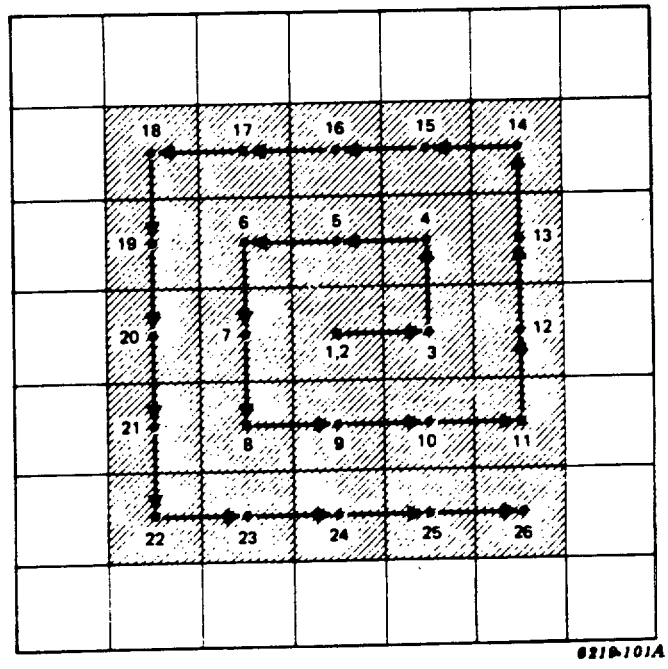


FIGURE 5. Scan pattern 5 x 5.

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3.3.1.4 Instantaneous FOV. The TCS shall incorporate two selectable Fields of View. The Narrow Field of View (NFOV) shall be 7.7 by 7.7 ±0.58 mrad (26.5 arc minutes ±2 arc minutes). The Wide Field of View (WFOV) shall be 24.7 by 24.7 ±2.9 mrad (85 arc minutes ±10 arc minutes).

3.3.1.5 Cone of vision. The cone of vision of the TCS shall be 30 ±1 degrees in azimuth and in elevation. The cone of vision shall be provided by a gimbal system within the TCA. Positioning of the gimbal system shall be controlled by signals/commands from the control-power supply unit.

3.3.1.6 Alignment.

3.3.1.6.1 Display centerline. A display reference centerline for test use shall be digitally generated within the TCS in the form of a Vertical Centerline (VCL) and a Horizontal Centerline (HCL). These signals, when combined with the test monitor video, shall define the center of the display raster with respect to time to ±0.7 percent of the sweep time. The Line of Sight (LOS) in either field of view is defined as the display raster center which is determined by the test display centerline.

3.3.1.6.2 Track window alignment. In the cage mode, the center of the track window shall be coincident with the center of the TV display raster within a tolerance of ±0.29 mrad (±1 arc minute) (1σ) in each axis, and shall be referred to the normal narrow field of view raster of 7.7 mrad (26.5 arc minutes) in height and width. The cage mode shall be the normal operating mode when not slaving, tracking, or using hand control inputs in the rate mode. In the cage mode, the slaving signals to the TCS shall not be used and the TCS gimbals shall be positioned so that they are boresighted with respect to the TCA mounting surface.

3.3.1.6.3 Bench alignment accuracy. The TCS, as delivered, shall be bench aligned, so that when caged the NFOV is within ±0.29 mrad (±1 arc minute) in both azimuth and elevation, with respect to the TCA mounting surface.

3.3.1.6.4 Bench boresight adjustments. The TCS shall include adjustments that provide for positioning of the narrow field Line of Sight (LOS) to within 0.07 mrad (±0.25 arc minute) (azimuth and elevation) of the TCA mounting surface. The roll adjustment shall be capable of positioning the NFOV LOS to the TCA mounting surface to within 1.45 mrad (±5 arc minutes) in roll.

3.3.1.7 Lines of sight. The narrow and wide-field lines of sight shall be coincident within 1.45 mrad (±5 arc minutes).

3.3.1.8 Sun sensor. The vidicon sensors (narrow and wide) shall be protected from sun illumination damage. The sun shutter mechanism shall be fail-safe in that power shall be required to keep the sun shutter open.

3.3.1.8.1 Sun shutter triggering levels. The sun sensor shall activate the protective shutter when the illumination level exceeds 20,000 ±5000 foot Lamberts (fL) from a 3000 Kelvin tungsten source.

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3.3.1.8.2 Sun sensor cone of view. The sun sensor shall have a cone of view with a 174 ±52 mrad (10 ±3 degrees) included angle.

3.3.1.8.3 Sun sensor shutter speed. The sun sensor shall close, after triggering, within 120 milliseconds.

3.3.1.9 Gimbal slewing dynamics. Stabilization shall be provided by a stabilized two-gimbal system. The maximum angular slew rate capability for the LOS shall be not less than 30 degrees/second. The maximum angular slew acceleration capability for the LOS shall be not less than 115 degrees/second^2. The TCS shall be capable of accepting LOS pointing commands or manual control stick commands as specified in 3.3.2.17.1 and 3.3.2.17.2. The stick position versus rate command shall be implemented as shown in Figure 6. Table I indicates control stick movement versus gimbal angle motion.

TABLE I. Stick motion versus gimbal angle motion.

Hand Control Stick Motion	Gimbal Angle, Cross-Hair Motion
Forward	Down
Aft	Up
Right	Right
Left	Left

3.3.1.10 Automatic video tracking target dynamics. The maximum video autotrack rate, in the WFOV and NFOV shall be not less than 87 mrad (5 degrees) per second. The maximum video autotrack acceleration, in the WFOV and NFOV, shall be not less than 72 mrad (4.1 degrees) per second^2.

3.3.1.11 Base Motion Isolation. Base motion isolation shall accommodate F-14 aircraft pitch and yaw rates of up to 3 rad (175 degrees) per second and angular accelerations up to 6.1 rad (350 degrees) per second^2.

3.3.1.12 Optical video tracker. The TCS shall provide automatic video tracking in either the WFOV or NFOV. The video tracking logic shall contain as a minimum:

a. Tracking of gated video
b. Adaptive tracking window size logic
c. Automatic contrast polarity selection logic

3.3.1.13 Static limiting resolution. The angular horizontal resolution of the complete system, when observing a stationary target (collimated at infinity) with the NFOV, shall be equal to or better than 34 μrad (7 arc seconds) when measured with a USAF 1951 resolution standard on Conrac Model QQA 14C-SP-1 video monitor or its equal. The wide field resolution, when measured under the same conditions, shall be equal to or better than 126 μrad (26 arc-seconds). (See Table II). The NFOV resolution, when measuring a target collimated at 7 nautical miles, shall be equal to or better than 34 μrad (7 arc seconds).

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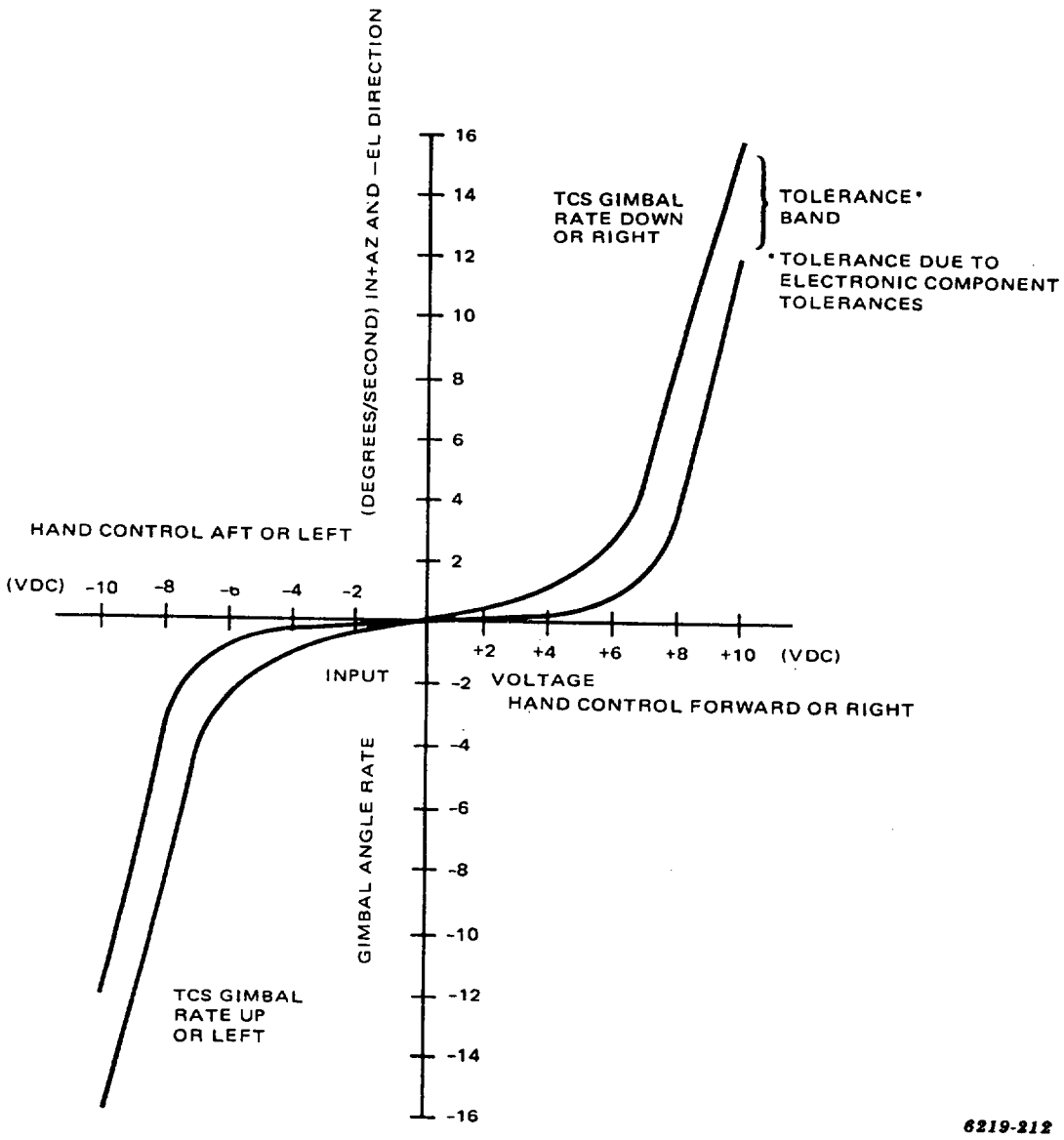


FIGURE 6. Gimbal rate for TCS versus input voltage.

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TABLE II. Static resolution and square wave response.

	Static Limiting Resolution		Square Wave Response		
			Cycles/Millirad		
	Millirad	Cycles/Millirad	6.5	13.0	19.5
NFOV	0.034	29.4	(100 TVL) 85%	(200 TVL) 55%	(300 TVL) 25%
WFOV	0.126	7.9	2.0 (100 TVL) 80%	4.0 (200 TVL) 45%	6.1 (300 TVL) 20%

3.3.1.14 Dynamic limiting resolution. The angular horizontal resolution of the complete system, when subjected to the vibration environment specified in 3.3.5.3, shall be equal to or better than 39 μrad (8 arc-seconds), when measured with a USAF 1951 resolution standard on a Conrac Model QQA 14C-SP-1 video monitor or its equal.

3.3.1.15 Square wave response. The modulation of the video signal, while observing a four-bar pattern, shall be greater than values listed in Table II. The modulation is determined by recording the peak-to-peak amplitude of the video signal (as observed on an oscilloscope) through the four-bar pattern, then calculate the percent modulation as follows:

$$\% \text{ modulation} = \frac{A_p}{M_o} \times 100$$

when A_p = recorded peak-to-peak amplitude of the modulated signal

M_o = the amplitude difference between the black and white level of the video signal measured at spatial frequency of approximately 25 TVL.

3.3.1.16 Gray scale rendition. The TCS video signal shall exhibit at least eight gray shades from a ten-shade scale when displayed on a Conrac Model QQA 14C-SP-1 video monitor or its equal. Gray shade steps are defined as follows:

Step	1	2	3	4	5	6	7	8	9	10
Percent Transmission	100	71	50	35	25	17.7	12.5	8.8	6.3	4.4

NOTE: 100 percent transmission is defined at a nominal 1500 fL scene brightness.

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3.3.1.17 Automatic light range. The TCS shall automatically maintain the video output peak white reference levels designated in 3.3.2.1 for scene brightness variations from 100 to 6000 foot-Lamberts. The automatic light range is based upon scene video and is independent of the display symbology signal component.

3.3.1.18 Scan failure protection. The TCS vidicon shall not incur damage or permanent degradation due to loss of sweep.

3.3.1.19 TCS geometric distortion. The total displacement of any point for both NFOV and WFOV shall be not greater than 2 percent of the full raster inside zone 1 and not greater than 4 percent of the full raster outside of zone 1. Zone 1 is a square area having a center coincident with the raster area and an area equal to one-half of the full raster area.

3.3.1.20 Television format. The TCS television format shall be characterized as follows:

a. 30 picture frames per second
b. 525 scan lines per frame
c. 2 fields per frame fixed interface
d. Unity (1:1) camera scan aspect ratio

3.3.1.21 Display symbology. Symbology information shall be contained in the four TCS video signal sources described in 3.3.2.1. The symbology shall be in the form of constant voltage amplitude pulses added to the scene during active raster scan periods. The following information shall be contained in the TCS display symbology signals:

a. Track window. The track window shall be as shown in Figures 7 and 8 and each corner representation shall be not less than two line widths in height with a nominal 1:1 aspect ratio.

b. TCS gimbal angle crosshair. The Gimbal Angle Crosshair (GACH) shall be as shown in Figures 7 and 8 and shall be 154 $\pm$24 raster lines in height and 17.8 $\pm$2.8 microseconds in width. (The line thickness shall be 2 $\pm$1 TV lines for the horizontal and 0.22 $\pm$0.12 μsec for the vertical.) The crosshair position in the raster display shall indicate the TCS LOS with respect to the FRL. The scaling shall be such that the total raster width represents $\pm$15° $\pm$10 percent in azimuth and the total raster height represents $\pm$15° $\pm$10 percent in elevation. In the TCS caged position, the crosshair center shall be in the timewise center of the display raster within $\pm$3 percent of the raster width and height.

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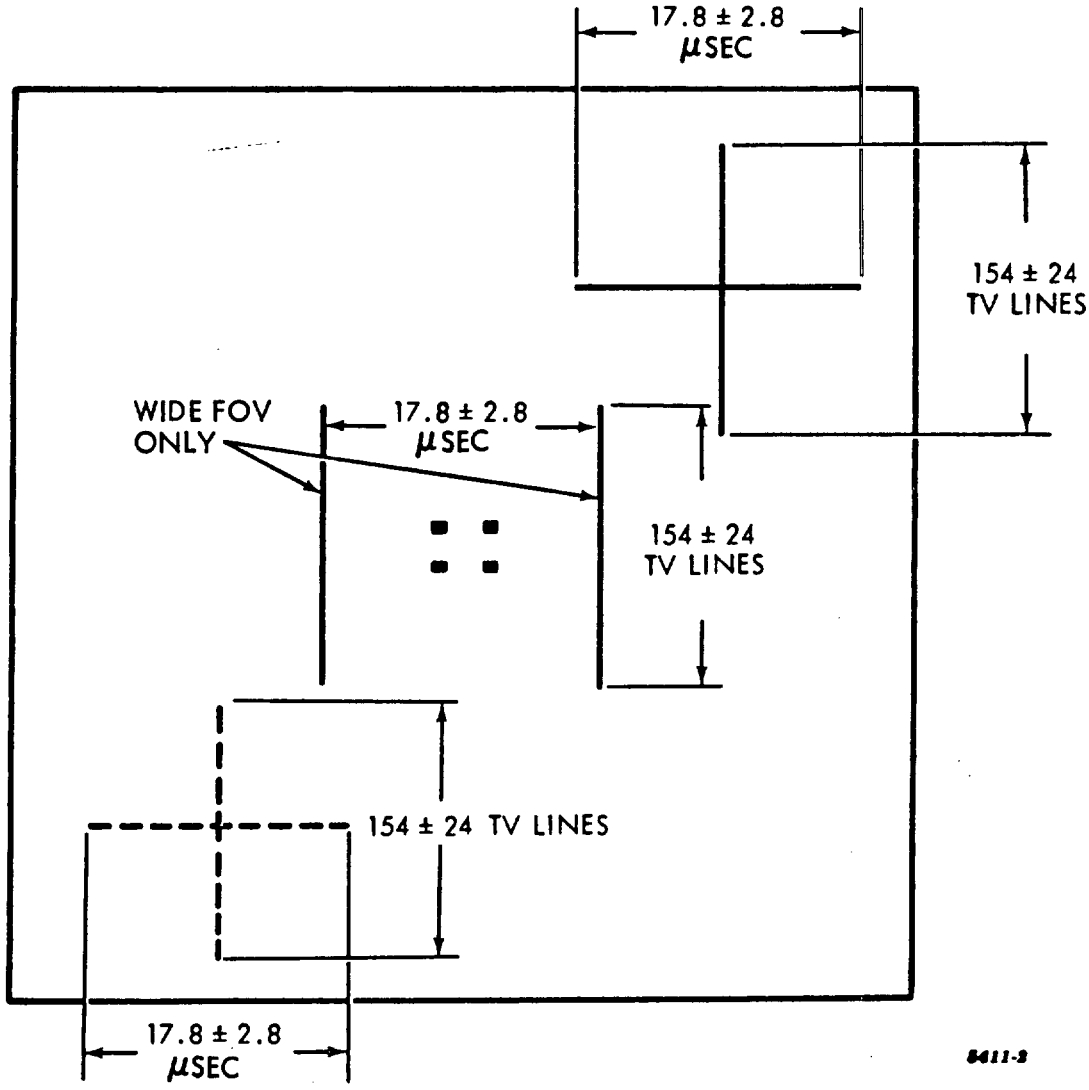


FIGURE 7. Video symbology of both fields of view.

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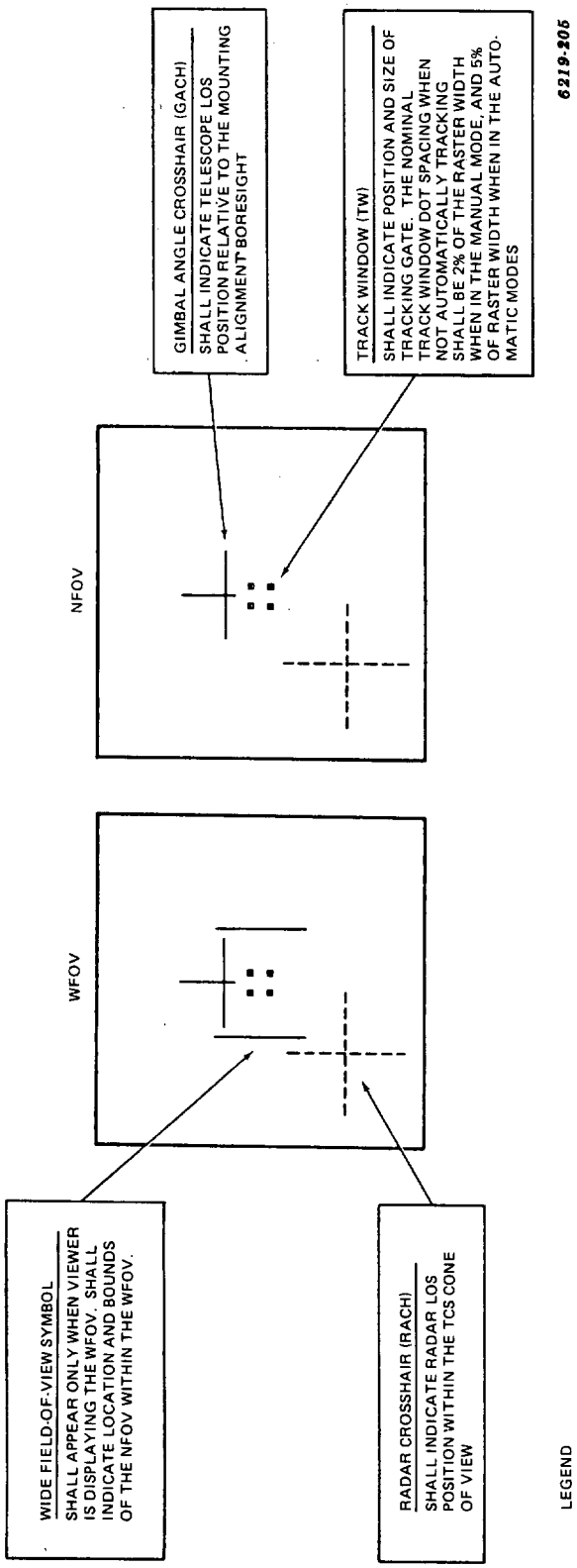


FIGURE 8. Symbology definition.

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c. Radar LOS position crosshair. The Radar Crosshair (RACH) symbol shall be as shown in Figures 7 and 8. The crosshair shall be a dash line cross with each dash consisting of a width of 0.4 ±0.1 microsecond and a height of 3 ±2 TV lines. (The line thickness shall be 2 ±1 TV lines for the horizontal and 0.22 ±0.12 μsec for the vertical.) The crosshair position in the raster shall indicate the radar LOS with respect to the FRL when the radar is tracking, or the commanded LOS when slaving to the AWG-9 computer. Scaling shall be the same as for the TCS gimbal angle crosshair.

d. Blanking of radar and TCS crosshairs. The radar and TCS crosshairs shall be blanked when over the track window to approximately twice the track window dimensions. The maximum dimension of the blanked area shall not exceed 15 percent of the total raster area.

e. Wide field of view symbol. The wide field of view symbol shall be as shown in Figures 7 and 8. It shall appear only when the viewer is displaying the wide field of view, and shall be 154 ±24 raster lines in height and 17.8 ±2.8 microseconds in width.

3.3.1.22 Angular accuracy. The TCS narrow field-of-view line-of-sight shall be positioned with respect to the TCS mounting provisions to an accuracy of ±1.5 milliradians (5 arc minutes) 2σ in both azimuth and elevation when slaved with errorless slaving signals, and with the radome refraction compensation circuit (paragraph 3.3.1.23) inhibited within the TCS. This accuracy shall apply throughout the 524 milliradian (30 degree) cone of vision of the TCS.

3.3.1.23 Radome refraction compensation. The TCS shall incorporate a radome refraction compensation circuit which adds a correction term to the slaving K direction component whenever the TCS is slaved in the minus elevation direction. The correction shall be effective in both the slave-to-radar and slave-to-computer modes. The correction polarity shall be such as to decrease the slaved "look-down" angle, and shall have a magnitude that is a function of the commanded minus elevation angle as shown in Figure 9. The corrected elevation angles versus commanded elevation angles shall be as specified in Table III. The correction circuit shall be capable of being inhibited during bench calibration and alignment of the TCS.

3.3.1.24 TCS reset. A TCS reset shall be initiated whenever the LOS limit switch is activated at gimbal limit, or whenever the hand control is enabled for TCS and the hand control action switch is placed in the half-action (HA) position and released. When TCS reset occurs and "Slave TCS" has been selected, the TCS shall reslave to the radar LOS if the radar is tracking, or to the AWG-9 computer if SLAC is true. If the conditions for reslaving are not met, the TCS shall go to its cage position.

3.3.2 TCS/F-14 interface. The following paragraphs define the characteristics of the interface between the TCS and F-14 aircraft. All angular position data is referenced to the TCS mounting surface.

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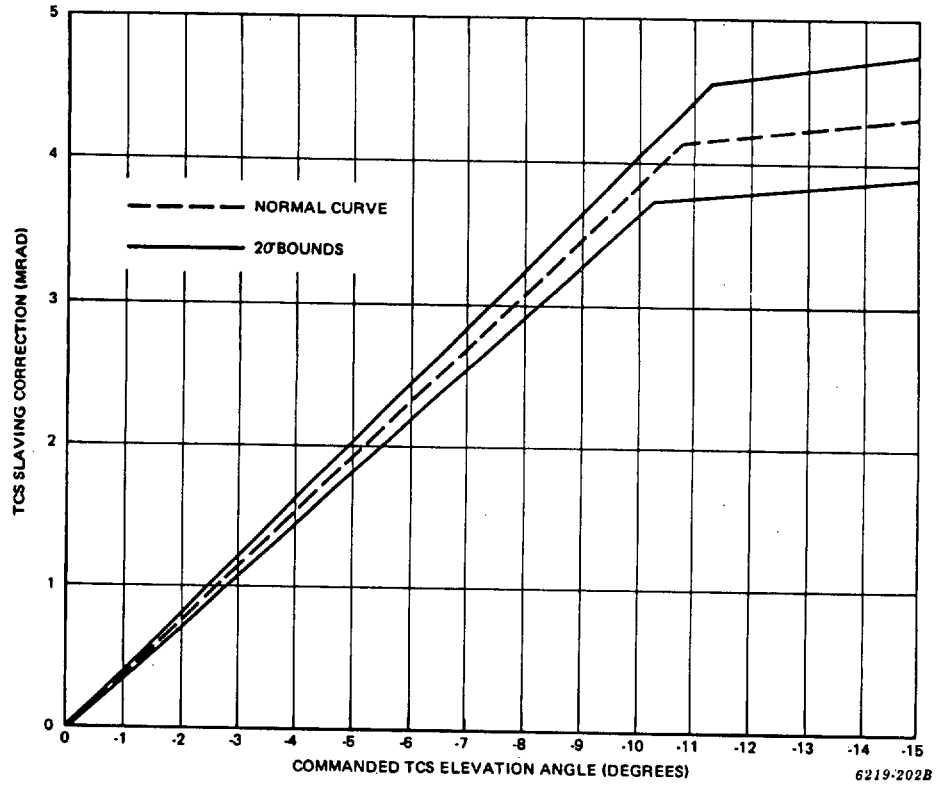


FIGURE 9. TCS elevation correction for radome refraction.

TABLE III. Radome Correction.

Commanded Elevation Angle (Deg)	Corrected Elevation Angle (Deg)	2σ Tolerance (Deg)
-1	-0.978	±0.0013
-2	-1.957	±0.0024
-3	-2.935	±0.0034
-4	-3.913	±0.0045
-5	-4.891	±0.0056
-6	-5.869	±0.0067
-7	-6.847	±0.0078
-8	-7.825	±0.0089
-9	-8.803	±0.010
-10	-9.781	±0.011
-11	-10.762	-0.016 +0.022
-12	-11.760	±0.023
-13	-12.757	±0.023
-14	-13.754	±0.024
-15	-14.752	±0.025

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3.3.2.1 Video outputs. The TCS shall provide four mutually isolated television video signal sources for the aircraft interface. Table IV specifies parameters that must be measured in accordance with 1, 2, and 3 below. Specifications for additional characteristics with respect to the four output signals follow the table.

1. All levels, except for the display symbology, are measured with respect to the blanking level.

2. All measurements shall be made at the TCS output terminals with the noted load impedances and do not take into consideration losses in the interface.

3. The black reference levels are specified as a maximum level with a scene illumination no less than 1500 foot-Lamberts and a vidicon ambient temperature no greater than 72° F.

TABLE IV. Output Destination

Parameter	DD	TID*	VTR	VDI*
Video Output Format	Clamped Composite	Clamped Composite	Clamped Composite	Clamped Only
Reference Figure No. Source Impedance	11 51Ω ±5%	11 91Ω ±5%	11 75Ω ±5%	10 91Ω ±5%
Load Impedance (Nominal)	50Ω	93Ω	75Ω	93Ω
Blanking Level (VDC)	0 ±0.1 V	0 ±0.1 V	0 ±0.1 V	0 ±0.1 V
White Reference Level (β)	2.2 Vdc ±10%	2.2 Vdc ±10%	1.8 Vdc ±10%	2.2 Vdc ±10%
Black Reference Level (Maximum)	0.16β	0.16β	0.16β	0.16β
Display Symbology	0.6 V PP ±10%	0.6 V PP ±10%	0.5 V PP ±10%	0.6 V PP ±10%
Sync Level (α)	-0.8 Vdc ±10%	-0.8 Vdc ±10%	-0.7 Vdc ±10%	---
Wiring, Triaxial, Nominal Impedance	50Ω	95Ω	75Ω	95Ω

*Separate Horizontal and Vertical sync signals shall be provided to the TID and VDI.

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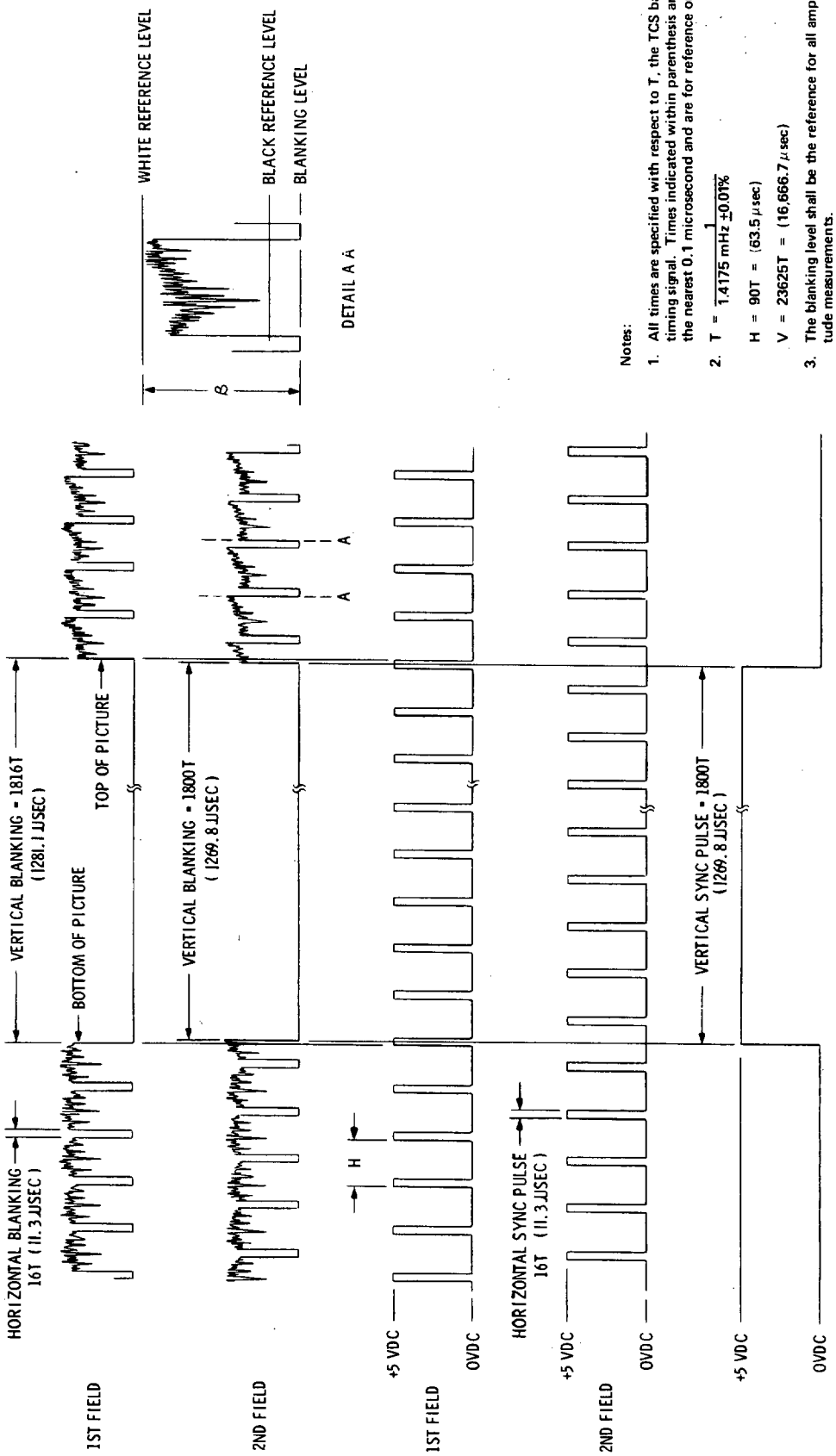
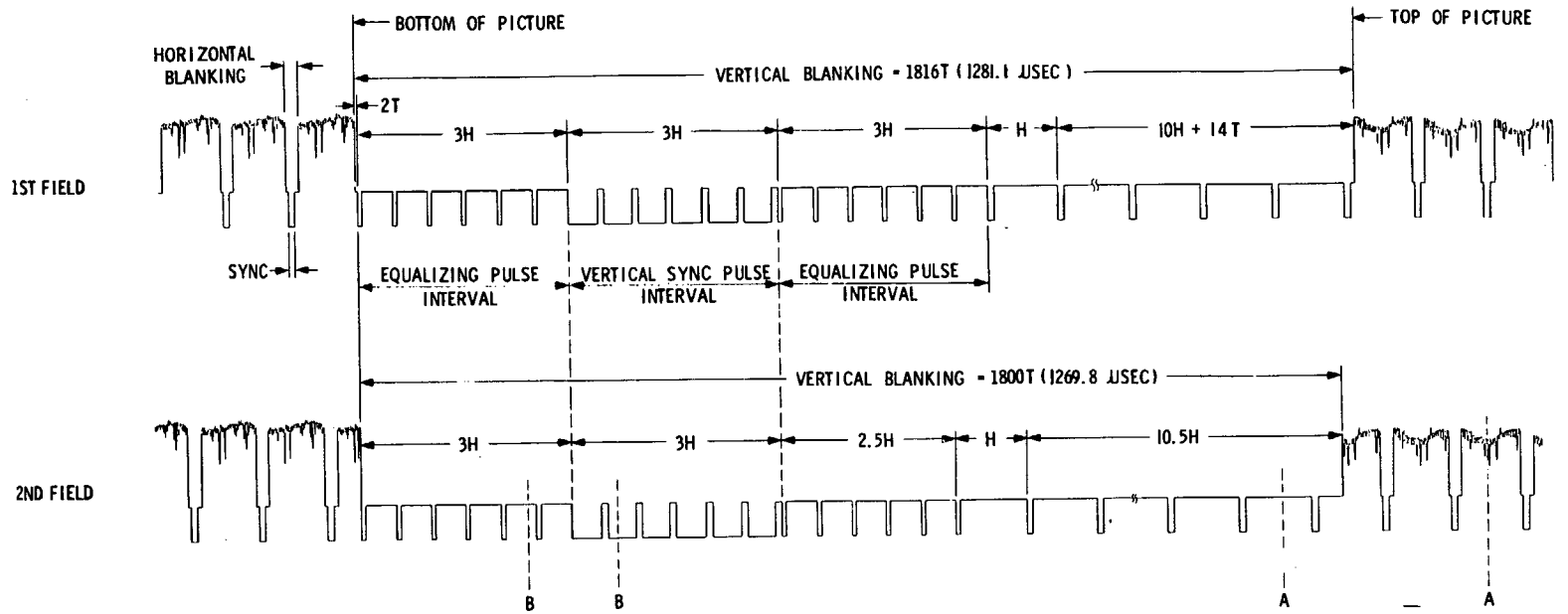
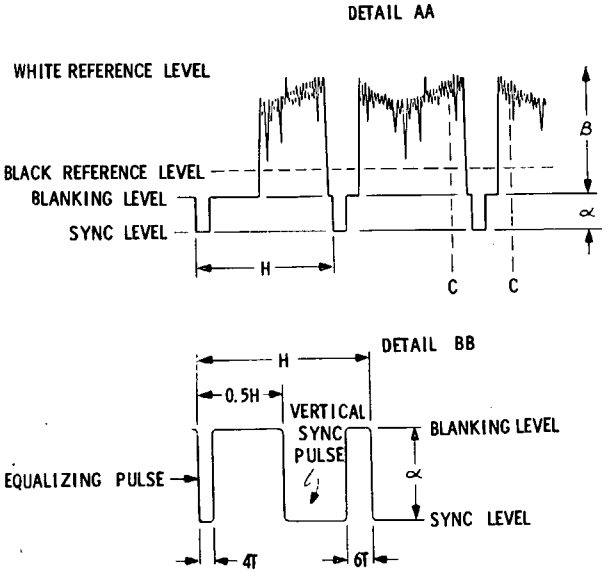
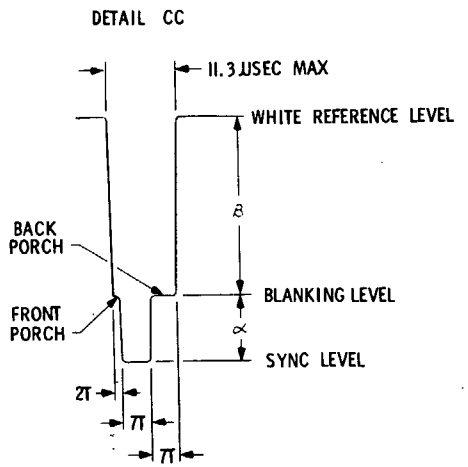


FIGURE 10. TCS clamped video format and external display synchronization signals.

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Notes:

1. All times are specified with respect to T, the TCS basic timing signal. Times indicated within parenthesis are to the nearest 0.1 microsecond and are for reference only.
2. $T = \frac{1}{1.4175\ \text{mHz} \pm 0.01\%}$

   $H = 90T = (63.5\ \mu\text{sec})$

   $V = 23625T = (16{,}666.7\ \mu\text{sec})$
3. The blanking level shall be the reference for all amplitude measurements.
4. The rise and decay time for all pulses shall not exceed 0.19 μsec when measured between the 0.1 and 0.9 amplitude excursion region.
5. The leading and trailing edge of the vertical blanking period shall be completed within 6.35 μsec.
6. The overshoot and the undershoot excursions of the horizontal blanking signal shall not exceed 0.02β at the beginning of the front porch and shall not exceed 0.05β at the end of the back porch.
7. The overshoot and the undershoot excursions of all sync pulses shall not exceed 0.5α at the beginning and shall not exceed 0.1α at the end.

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FIGURE 11. TCS composite video format.

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a.	Shading horizontal	Less than 35 percent over the center 80 percent of the raster
b.	Horizontal period	63.49 ±0.01 microseconds
c.	Vertical period	16,666.7 ±1.7 microseconds
d.	Polarity	Black - negative
e.	Bandwidth	16 Hz to 12 MHz (NFOV), 16 Hz to 8 MHz (WFOV), at 3 dB down
f.	Square Wave Response	See Table II (Static Resolution and Square Wave Response)
g.	Common mode noise	40 dB less than the signal at the TCS outputs
h.	Blanking duration	Horizontal: 11.29 ±0.01 microseconds Vertical 1st field: 1281.13 ±0.13 microseconds Vertical 2nd field: 1269.84 ±0.13 microseconds
i.	Noise in blanking	Shall not exceed 7 percent of white level

3.3.2.2 Horizontal sync signal. The TCS shall provide two separate horizontal sync outputs to the interface, one output shall be for the AN/AWG-9 TID and the other output for the VDI display. Each signal shall provide the necessary drive voltage and timing to lock the TV picture at the horizontal frequency and shall have the following characteristics measured across a 93-ohm, nominal, termination at the output of the TCS system. (Reference Figure 10 for time/phase relationship.)

a.	Type signal	Pulse
b.	Polarity	Positive
c.	Rise time	Equal to or less than 190 nanoseconds
d.	Fall time	Equal to or less than 190 nanoseconds
e.	Duration	11.29 ±0.01 microseconds
f.	Period	63.49 ±0.01 microseconds
g.	Amplitude	+5 ±0.5 volt is on state; $0.0\ ^{+0.4}_{-0.0}$ volt is off state
h.	Stability	As specified in EIA Standard RS-170

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i.	Noise	Shall not exceed 7 percent of sync amplitude
j.	Wiring	Triax, 95 ±5 ohms

3.3.2.3 Vertical sync signal. The TCS shall provide two separate vertical sync outputs to the interface, one output shall be for the AN/AWG-9 TID and the other output for the VDI display. Each signal shall provide the necessary drive voltage and timing to lock the TV picture at the vertical frequency. It shall have the following characteristics measured across a 93-ohm ±10 percent termination at the output of the TCS system. (Reference Figure 10 for time/phase relationship.)

a.	Type signal	Pulse
b.	Polarity	Positive
c.	Rise time	Equal to or less than 6.3 microseconds
d.	Fall time	Equal to or less than 6.3 microseconds
e.	Duration	1269.84 ±0.13 microseconds
f.	Period	16,666.7 ±1.7 microseconds
g.	Amplitude	+5 ±0.5 volts is on state; $0.0\ ^{+0.4}_{-0.0}$ volt is off state
h.	Stability	As specified in EIA Standard RS-170
i.	Noise	Shall not exceed 7 percent of sync amplitude
j.	Wiring	Triax, 95 ±5 ohms

3.3.2.4 TCS onboard signal. This signal shall indicate to the AN/AWG-9 radar that the TCS is on board the aircraft and shall have the characteristics specified below. This signal shall be provided to the interface for the AN/AWG-9 081 unit which will provide this signal to the computer subsystem via the serial data interface when terminated with a 2.7-kilohm ±2 percent load.

a.	Type signal	Switched dc
b.	Voltage amplitude	+12 ±1 Vdc when TCS is onboard and has power on, and open circuit when TV is not aboard.
c.	Wiring	Unshielded

3.3.2.5 TCS temperature warning signal. This signal shall indicate when the TCS is operating in an over-temperature condition. The over-temperature signal shall be generated when the internal temperature of the TCA exceeds 140° F ±3.5 percent. This signal shall be provided to the interface for energizing an indicator on the hand control unit. Signal characteristics are described in 3.3.2.21.

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3.3.2.6 TCS elevation gimbal. This signal shall provide the elevation gimbal angle signal of the TCS to the interface for the Detailed Data Display (DDD). The signal shall have the following characteristics:

a.	Type signal	Dc analog
b.	Scale factor	61 ±4.4 mrad (3.5 ±0.25 degrees) per volt (minus voltage up and plus voltage down)
c.	Load impedance	37 kilohms or greater

3.3.2.7 TCS azimuth gimbal. This signal shall provide the azimuth gimbal angle of the TCS to the interface for the DDD. The signal shall have the following characteristics:

a.	Type signal	Dc analog
b.	Scale factor	61 ±4.4 mrad, (3.5 ±0.25 degrees) per volt (minus voltage to the right, plus voltage to the left)
c.	Load impedance	37 kilohms or greater
d.	Wiring	Shielded

3.3.2.8 TCS optical position to radar. The TCS optical position signal shall be provided to the interface for the AN/AWG-9 radar subsystem in the form of three direction cosine signals (in order that the radar antenna may be slaved to the TCS optical position) (Figure 12). The three direction cosine signals and their characteristics are defined as follows when terminated in a load 150-kilohm or greater impedance. [The accuracy of the angle defined by the i, j, -k signals shall be 12 mrad (41.25 arc minutes) 2σ in elevation and 6 mrad (20.6 arc minutes) 2σ in azimuth.]:

i TCS Track

a.	Track signal	Ac analog
b.	Voltage amplitude	7.5 ±10% volts rms cosine AZ cosine EL
c.	Frequency	400 ±12 Hz
d.	Source impedance	Less than 800 ohms
e.	Wiring	Shielded
f.	Remarks	AZ and EL are actual TCS gimbal angles

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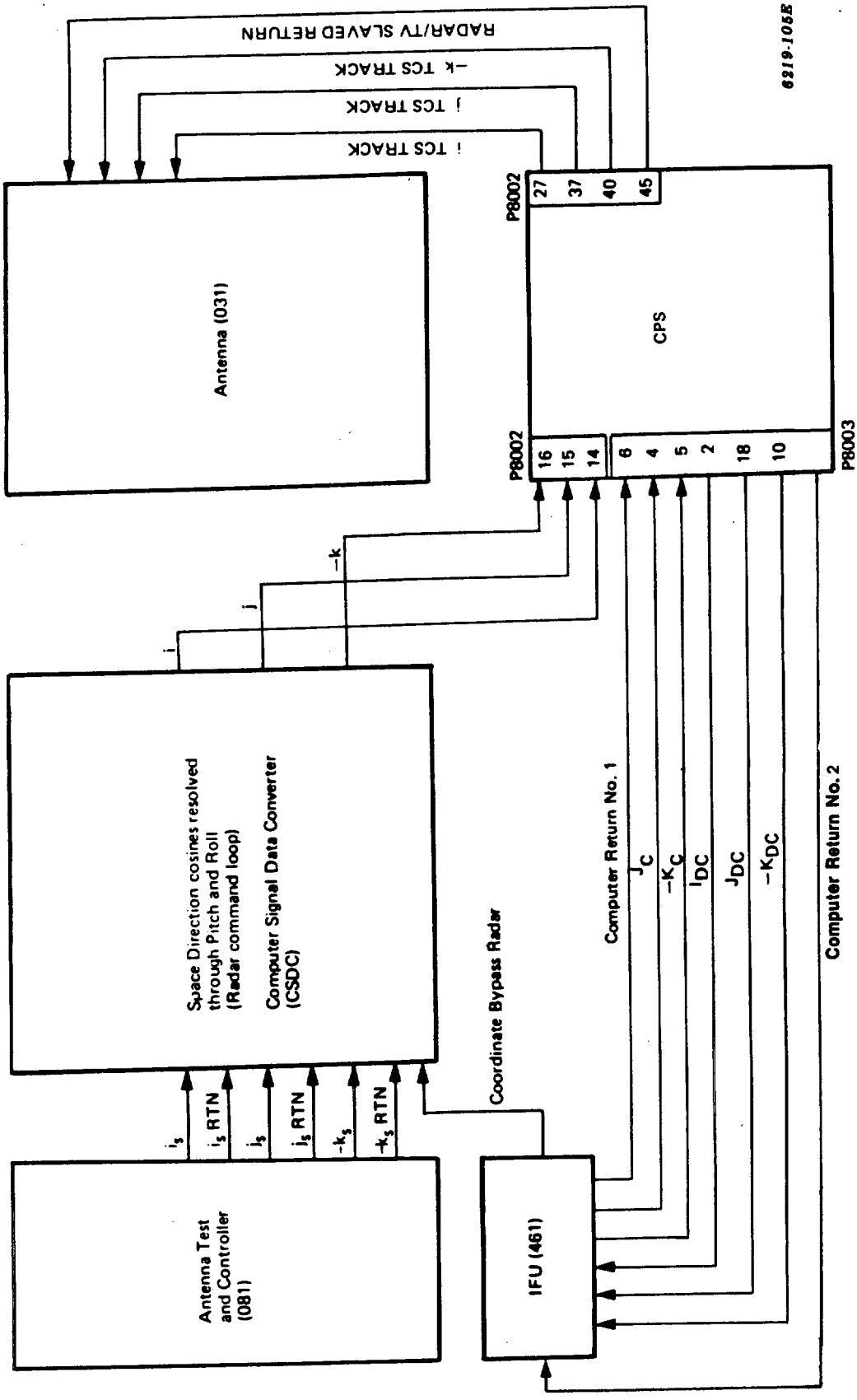


FIGURE 12. Coordinate transformation channel.

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j TCS Track

a.	Type signal	Ac analog
b.	Voltage amplitude	7.5 ±10% volts rms sine AZ cosine EL
c.	Frequency	400 ±12 Hz
d.	Source impedance	Less than 800 ohms
e.	Wiring	Shielded
f.	Remarks	AZ and EL are actual TCS gimbal angles. Output null voltage at AZ=0, consisting of quadrature and harmonics, shall be less than 50 millivolts rms.

-k TCS Track

a.	Type signal	Ac analog
b.	Voltage amplitude	7.5 ±10% volts rms sine EL
c.	Frequency	400 ±12 Hz
d.	Source impedance	Less than 800 ohms
e.	Wiring	Shielded
f.	Remarks	EL is actual TCS gimbal angle. Output null voltage at EL=0, consisting of quadrature and harmonics, shall be less than 50 millivolts rms.

Radar Slaved Return

a.	Type signal	Ac analog return
b.	Signal function	Signal return

3.3.2.9 TCS optical position to AN/AWG-9 computer. The TCS optical position shall be provided to the interface for the AN/AWG-9 computer in the form of three dc direction cosine signals. These signals shall be available during all modes of TCS operation. The three dc direction cosine signals and their characteristics shall be as defined below. These signals shall track the AWG-9 +10 Vdc analog reference defined in 3.3.2.13. The accuracy of the angle defined by the I, J, -K signals shall be 2 mrad (6.9 arc minutes) 2σ in azimuth and elevation:

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I_{DC}

a.	Type signal	Dc analog
b.	Voltage amplitude	10 cosine AZ cosine EL (Vdc)
c.	Source impedance	Less than 20 ohms
d.	Load impedance	Greater than 20 kilohms
e.	Wiring	Shielded
f.	Remarks	AZ and EL are actual TCS gimbal angles

J_{DC}

a.	Type signal	Dc analog
b.	Voltage amplitude	40 sine AZ cosine EL (Vdc)
c.	Source impedance	Less than 20 ohms
d.	Load impedance	Greater than 20 kilohms
e.	Wiring	Shielded
f.	Remarks	AZ and EL are actual TCS gimbal angles

$-K_{DC}$

a.	Type signal	Dc analog
b.	Voltage amplitude	40 sine EL (Vdc)
c.	Source impedance	Less than 20 ohms
d.	Load impedance	Greater than 20 kilohms
e.	Wiring	Shielded
f.	Remarks	EL is actual TCS gimbal angle

3.3.2.10 TCS gimbal angle from AN/AWG-9 computer. The AN/AWG-9 computer will provide AZ and EL directional information to the TCS in order to acquire and lock on a target. In the Auto-Acquisition mode, the target must appear in the auto-acquisition window which encompasses as a minimum the center 50 percent of the field of view. This requires that the TCS be slaved to the target with an accuracy of $\pm$2.72 millirad in each axis. The two dc directional cosine signals shall have the following characteristics when terminated in a 200-kilohm $\pm$1 percent load:

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J_C

a.	Type signal	Dc analog
b.	Voltage amplitude	40 cosine EL sine AZ (Vdc)
c.	Source impedance	Less than 20 ohms
d.	Wiring	Shielded
e.	Remarks	AZ and EL are commanded TCS gimbal angles. Maximum signal amplitude is equal to the ±10 Vdc AWG-9 analog reference.

$-K_C$

a.	Type signal	Dc analog
b.	Voltage amplitude	40 sine EL (Vdc)
c.	Source impedance	Less than 20 ohms
d.	Wiring	Shielded
f.	Remarks	EL is the commanded TCS elevation gimbal angle. Maximum signal amplitude is equal to the ±10 Vdc AWG-9 analog reference.

3.3.2.11 Computer return #1. This line connects the computer interface unit (962 rack) ground to the TCS, and shall serve as the J_C and $-K_C$ signal return.

3.3.2.12 SPATS (Search Pattern Size). The TCS shall require a logic level signal to select between a 3 x 3 search pattern and a 5 x 5 search pattern.

a.	Type signal	Digital type A avionics (see Appendix B)
b.	Logic "One"	Commands a 3 x 3 search pattern when AUTO-SEARCH has been selected on the SENSOR CONTROL PANEL "ACQUISITION" SWITCH.
c.	Logic "Zero"	Commands a 5 x 5 search pattern when AUTO-SEARCH has been selected on the SENSOR CONTROL PANEL "ACQUISITION" SWITCH.

3.3.2.13 +10 Vdc analog reference. A +10 Vdc analog reference is supplied from the AN/AWG-9. This voltage shall be used to scale the TCS generated I, J, K set to be compatible with the slaving inputs J_C and $-K_C$. It shall also be used to scale the demodulated 400 Hz slaving signals j_i and $-k_i$ from the CSDC to be compatible with the TCS generated I, J, K set.

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a.	Type signal	Dc
b.	Voltage amplitude	+10 V ±1 percent
c.	Source impedance	20 ohms max.
d.	Load impedance	200 kilohms ±1 percent
e.	Remarks	This voltage tracks the positive portion of the +10 Vdc analog reference used in the A/D and D/A converter in the AN/AWG-9 computer interface unit (962 rack).

3.3.2.14 TCS track valid. This signal shall indicate that the TCS is automatically tracking a target. This signal shall be provided to the interface on the existing "On Target" (OT) indicate line to the DDD unit and on the existing "track valid" line to the computer Interface Unit (IFU) when the TCS is onboard.

a.	Type signal	Switched dc
b.	Voltage amplitude	+12 ±1 Vdc shall indicate tracking, 0 ±1 Vdc indicates not tracking
c.	Load impedance	1.3 kilohms min in the DDD unit and 1.5 kilohms ±30 percent in the computer interface unit
d.	Wiring	Shielded

3.3.2.15 Computer return #2. This line connects the computer interface unit (962 rack) ground to the TCS, and shall be used in the TCS so that the TCS outputs I_{DC}, J_{DC} and $-K_{DC}$ are referenced to the computer interface unit ground.

3.3.2.16 TCS power interlock signal. The TCS shall provide a power interlock signal to the F-14 power control distribution system when the AN/AWG-9 is in STANDBY or ON and the TCS power switch is placed to STBY or ON and the CPS and TCA are installed in the aircraft. This signal commands the F-14 power distribution and control system to provide 3-phase, 400-Hz, 115 Vac power to the TCS subsystem. This signal shall have the following characteristics:

a.	Type signal	Switched dc
b.	Voltage Amplitude	28 Vdc +5.2, -2.9 Vdc or open circuit
c.	Load Current	125 mA max
d.	Remarks	The source of this 28 Vdc signal is the TCS Standby Select signal (see 3.3.2.17.10).

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3.3.2.17 TCS control inputs from AN/AWG-9 and F-14

3.3.2.17.1 TCS hand control X. The TCS shall respond to the hand control X (Azimuth Positioning) signal from the AN/AWG-9 hand control unit. This signal shall have the following characteristics when terminated with a load of 30 kilohms or greater impedance:

a.	Type signal	Dc analog
b.	Voltage amplitude	+10 to -10 Vdc ±3%, positive voltage with hand control stick to the right
c.	Wiring	Shielded

3.3.2.17.2 TCS hand control Y. The TCS shall respond to the hand control Y (Elevation Positioning) signals from the AN/AWG-9 hand control unit. This signal shall have the following characteristics when terminated with a load of 30 kilohms or greater impedance:

a.	Type signal	Dc analog
b.	Voltage amplitude	+10 to -10 Vdc ±3 percent, positive voltage with hand control stick forward
c.	Wiring	Shielded

3.3.2.17.3 Half action. The TCS shall respond to the hand control half action signal from the AN/AWG-9 hand control unit. When the Stick has been selected for TV control and True signal (as defined below) is received, the TCS shall switch to a manual rate mode. This signal shall have the following characteristics when terminated with a 5-kilohm ±10 percent load:

a.	Type signal	Switched dc
b.	Voltage amplitude	Open circuit for false state, +12 ±0.5 Vdc for true state
c.	Wiring	Unshielded

3.3.2.17.4 Full action. The TCS shall respond to the hand control full action signal from the AN/AWG-9 hand control unit. When the Stick has been selected for TV control and a True signal (as defined below) is received, the TCS shall initiate video tracking sequences. This signal shall have the following characteristics when terminated with a 5-kilohm ±10 percent load:

a.	Type signal	Switched dc
b.	Voltage amplitude	Open circuit for false state, +12 Vdc ±0.5 Vdc for true state

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c.	Wiring	Unshielded
d.	Remarks	Half Action held true.

3.3.2.17.5 TCS power on select. This signal shall be used by the TCS system to apply ac power to the appropriate module in the TCS subsystem. The signal shall have the following characteristics when terminated with a 125-ohm minimum load impedance:

a.	Type signal	Switched dc
b.	Voltage amplitude	+28 Vdc, +5.2, -2.9 Vdc or open; +28 Vdc is the true state
c.	Wiring	Unshielded
d.	Remarks	The 28 Vdc is provided by the AN/AWG-9

3.3.2.17.6 AUTO-ACQ. When this signal is true, the TCS automatic acquisition mode shall be selected. The signal shall have the following characteristics when terminated with a 2-kilohm ±10 percent load:

a.	Type signal	Switched dc
b.	Voltage amplitude	+12 Vdc ±0.5 Vdc or open circuit. +12 Vdc is the true state
c.	Wiring	Unshielded
d.	Remarks	The 12 Vdc is provided by the AN/AWG-9.

3.3.2.17.7 AUTO-SEARCH. When this signal is true, the automatic-acquisition with Search Mode shall be selected. The signal shall have the following characteristics when terminated with a 2-kilohm ±10 percent load:

a.	Type signal	Switched dc
b.	Voltage amplitude	+12 Vdc ±0.5 Vdc or open circuit. +12 Vdc is the true state
c.	Wiring	Unshielded
d.	Remarks	The +12 Vdc is provided by the AN/AWG-9.

3.3.2.17.8 RDR/TCS trim azimuth. The azimuth boresight trim provides an adjustable electrical error correction which can be used to modify the demodulated and normalized j component of the ac radar slaving signal. This signal shall act to trim out azimuth misalignment between the radar and slaved TCS lines of sight at boresight position. It will have the following characteristics when terminated with a 100-kilohm ±10 percent load:

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a.	Type signal	Dc analog
b.	Voltage amplitude	-12 ±0.3 Vdc to +12 ±0.3 Vdc continuously adjustable to 0.1 V or less increments
c.	Nominal range of adjustment	±1.95°
d.	Wiring	Shielded
e.	Polarity	(+) moves head to left

3.3.2.17.9 RDR/TCS trim elevation. The elevation boresight trim provides an adjustable electrical error correction which can be used to modify the demodulated and normalized k component of the ac radar slaving signal. This signal shall act to trim out elevation misalignment between the radar and slaved TCS lines of sight at boresight position. It will have the following characteristics when terminated with a 100-kilohm ±10 percent load:

a.	Type signal	Dc analog
b.	Voltage amplitude	-12 ±0.3 Vdc to +12 ±0.3 Vdc continuously adjustable to 0.1 V or less increments
c.	Nominal range of adjustment	±1.95°
d.	Wiring	Shielded
e.	Polarity	(+) moves head up

3.3.2.17.10 TCS standby select. The TCS Standby Select signal is supplied from the Hand Control Unit TV power switch when either in the STBY or in the ON position. This signal shall be used by the TCS to operate relays in both the CPS and TCA and to provide the voltage source for the TCS Power Interlock Signal. The TCS shall respond to a signal having the following characteristics when terminated with a 76-ohm minimum impedance:

a.	Type signal	Switched dc
b.	Voltage amplitude	+28 Vdc, +5.2, -2.9 Vdc or open; +28 Vdc is the true state
c.	Wiring	Unshielded
d.	Remarks	See 3.3.2.16 for additional information. TCS Standby Select signal is TRUE whenever TCS On Select is TRUE

3.3.2.17.11 TCS slave select. The TCS shall respond to a slave select signal provided to the interface from the sensor control panel. The SLAVE switch

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in the TCS position shall command the TCS to slave to the radar when the radar is tracking or to the AWG-9 computer when the SLAC signal (see 3.3.2.17.16) is true. This signal will have the following characteristics when terminated with a 10-kilohm ±10 percent load:

a. Type signal	Switched dc
b. Voltage amplitude	+12 ±0.5 Vdc or open +12 Vdc is TCS slave select
c. Wiring	Unshielded.

3.3.2.17.12 Radar tracking (RTG). The TCS shall receive a signal that indicates that the radar is tracking a target. This signal will have the following characteristics:

a. Type signal	Digital type A avionics (see Appendix B)
b. Wiring	Shielded

3.3.2.17.13 104 Vrms, 400 Hz REF. This reference voltage is supplied from the AN/AWG-9 Regulated Power Supply (601 unit). The TCS shall use this voltage for resolver excitation and as the demodulator reference for TCS resolver outputs and slaving signals from the radar. The TCS shall respond to signals having the following characteristics when terminated with a load of 1.7 kilohms minimum impedance:

a. Type signal	Ac
b. Voltage amplitude	104 Vrms ±3 percent
c. Peak-to-peak amplitude modulation distortion	Less than 3 percent about the steady state reference level measured over a bandwidth from 1 to 175 Hz
d. Total harmonic distortion	Less than 6 percent
e. Ac reference phase lock	Phase locked to the fundamental frequency of the input line, Phase A to within ±5 degrees of a fixed phase shift in the power subsystem. The fixed phase shift shall be from 0 to 25 degrees.
f. Frequency	400 ±12 Hz
g. Signal return	104 V 400 Hz reference return

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h.	Wiring	Twisted shielded pair
i.	Remarks	104 Vrms, 400 Hz return is not connected to ground.

3.3.2.17.14 Rear video select. This signal shall be used by the TCS to select narrow or wide field-of-view video sent to the rear cockpit displays, and to the Sensor Control Panel for the video tape recorder.

a.	Type signal	Switched dc
b.	Voltage amplitude	+12 ±0.5 Vdc or open circuit +12 Vdc selects NFOV Open circuit selects WFOV

3.3.2.17.15 Front Video Select (Option). To utilize this option, a jumper on Video Output card 339735 must be removed. This signal shall be used by the TCS to select narrow or wide field-of-view video sent to the Vertical Display Indicator (VDI) in the front cockpit.

a.	Type signal	Dc voltage supplied by the TCS is grounded or left open
b.	Voltage amplitude	5 ±0.2 Vdc from a 1 K ±25 percent source Open circuit selects NFOV Ground selects WFOV

3.3.2.17.16 SLAC (Slave to Computer). The TCS shall slave to computer generated LOS information J_c and $-K_c$ when SLAC is high, and the SLAVE switch on the sensor control panel is in the TCS position.

a.	Type signal	Digital type A avionics (see Appendix B)
b.	Logic "one"	Commands TCS to slave to computer when the AWG-9 SENSOR CONTROL PANEL SLAVE SWITCH is in the TCS position
c.	Logic "zero"	TCS does not slave to computer

3.3.2.18 TCS analog inputs from Computer Signal Data Converters (CSDC). The CSDC will provide, to the TCS, three directional cosine signals which define the directional coordinates of the radar antenna. In the Auto-Acquisition mode, the target must appear within the auto-acquisition window, which encompasses as a minimum, the center 50 percent of the field of view. This requires that the TCS be slaved to the target with an accuracy of ±2.72 mrad in each axis. The directional cosine signals shall have the following characteristics.

3.3.2.18.1 Antenna position. The TCS shall respond to the three-directional cosine signals from the CSDC when the TCS slave mode is selected and the radar is tracking a target (Figure 12). The unit vector for the three

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direction cosine signals is defined by U = 7.5 Vrms (Secant Epsilon Sub s) Absolute Cosine Multiplier (ACM) where Epsilon Sub s (ϵ_s) is the radar elevation angle measured from the local horizontal (up is positive).

ACM = 1.000,	$0° \leq \lvert\epsilon_s\rvert \leq 29.6°$,	$7.500 \leq U \leq 8.626$
ACM = 0.782,	$29.6° \leq \lvert\epsilon_s\rvert \leq 51.3°$,	$6.745 \leq U \leq 9.380$
ACM = 0.488,	$51.3° \leq \lvert\epsilon_s\rvert \leq 63.9°$,	$5.854 \leq U \leq 8.319$
ACM = 0.247,	$63.9° \leq \lvert\epsilon_s\rvert \leq 76°$,	$4.211 \leq U \leq 7.657$

U is expressed in Vrms. ϵ_s may have a tolerance of ± 2 degrees with respect to the points at which ACM changes. The characteristics of the signals are defined in terms of i_s, j_s, $-k_s$ (space coordinates) and θ and ϕ. These terms are defined as follows, where Eta Sub s (η_s) is the radar azimuth angle measured from the i_s vector to the projection of unit vector on the i_s j_s plane (Right is positive):

i_s = U Cos ϵ_s Cos η_s = 7.5 Vrms Cos η_s ACM

j_s = U Cos ϵ_s Sin η_s = 7.5 Vrms Sin η_s ACM

$-k_s$ = U Sin ϵ_s = 7.5 Vrms Tan ϵ_s ACM

θ = Aircraft pitch angle (Up is positive)

ϕ = Aircraft roll angle (Right wing down is positive)

The characteristics of the signals are defined as follows:

a. i

(1)	Type signal	Ac analog
(2)	Voltage amplitude	i_s Cos θ - k_s Sin θ
(3)	Frequency	400 $\pm$12 Hz
(4)	Source Impedance	Less than 800 ohms
(5)	Load Impedance	499 kilohms $\pm$1 percent
(6)	Wiring	Shielded

b. j

(1)	Type signal	Ac analog
(2)	Voltage amplitude	i_s Sin θ Sin ϕ + j_s Cos ϕ + k_s Cos θ Sin ϕ
(3)	Frequency	400 $\pm$12 Hz

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(4)	Source Impedance	Less than 800 ohms
(5)	Load Impedance	499 kilohms $\pm$1 percent
(6)	Wiring	Shielded

c. -k

(1)	Type signal	Ac analog
(2)	Voltage amplitude	$-i_s \text{ Sin } \theta \text{ Cos } \phi + j_s \text{ Sin } \phi - k_s \text{ Cos } \theta \text{ Cos } \phi$
(3)	Frequency	400 $\pm$12 Hz
(4)	Source Impedance	Less than 800 ohms
(5)	Load Impedance	499 kilohms $\pm$1 percent
(6)	Wiring	Shielded

NOTE: The i, j, -k voltage amplitudes may also be expressed in terms of the TCS Elevation and Azimuth gimbals (ϵ_g and η_g respectively) and ϵ_s and ACM.

$$i = 7.5 \text{ Vrms Cos } \epsilon_g \text{ Cos } \eta_g \text{ Sec } \epsilon_s \text{ ACM}$$

$$j = 7.5 \text{ Vrms Cos } \epsilon_g \text{ Sin } \eta_g \text{ Sec } \epsilon_s \text{ ACM}$$

$$-k = 7.5 \text{ Vrms Sin } \epsilon_g \text{ Sec } \epsilon_s \text{ ACM}$$

3.3.2.19 TCS coolant input from F-14. The TCS shall perform as specified herein when provided with liquid coolant under the following conditions to the TCS system:

a.	Coolant	Coolanol 25 (Monsanto) or Oronite Flo-Cool 180 (Chevron)
b.	Inlet Temperature	50-85° F
c.	Flow rate	0.70 gal/min max TCA - 0.40 gal/min max CPS - 0.30 gal/min max
d.	Inlet Pressure Discharge Pressure	170 psig max (@ rated flow) 35 psia min (@ rated flow)
e.	Unit Pressure Drop	75 psi at specified flowrates and 65° coolant temperature.

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3.3.2.20 Input electrical power.

3.3.2.20.1 Operating power. The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as specified in MIL-STD-704, Category B. The power required shall be not greater than 600 VA when supplied with three-phase ac power at 115/200 volts in accordance with MIL-STD-704, Category B.

3.3.2.20.2 Lighting power. Input power for lighting shall not be required.

3.3.2.20.3 Degraded performance. Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.2.20.4 Standby power. A control shall be provided to activate the TCS internal blowers and heaters while the remainder of the system is off. Except for the associated heaters and blowers, voltages shall not be applied to other parts of the equipment. The standby power shall be not greater than 400 VA.

3.3.2.21 TCS temperature warning signal. This signal shall indicate that the TCS is operating in an over-temperature condition. The over-temperature signal shall be generated when the internal temperature of the TCA exceeds 140° F ±3.5 percent. This signal shall be provided to the interface for the hand control. The signal shall have the following characteristics when terminated with a 6.8-kilohm load to -12 Vdc.

a.	Type signal	Switched dc
b.	Voltage amplitude	+12 ±1 Vdc indicates over temperature and open circuit indicates normal temperature
c.	Wiring	Unshielded

3.3.2.22 Cabling and connections.

3.3.2.22.1 Cables and connectors. The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400 or MIL-C-83723.

3.3.2.22.2 Interconnecting cabling. The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.3 Reliability. The contractor shall define and conduct a reliability program in accordance with MIL-STD-785 and the additional requirements of this document. The written Reliability Program Plan shall be the master planning and control document for the contractor's reliability program. Acceptance of

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the contractor's articles will not be made unless the reliability program plan, as approved, is maintained and implemented by the contractor. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.3.1 Operational stability. The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 580 hours without the necessity for readjustment of any controls (with the exception of the vidicon grid voltage adjustments) which are inaccessible to the operator during normal use.

3.3.3.2 Operating life. The equipment shall have an indefinite operating life with minimal servicing and replacement of parts.

3.3.3.3 Reliability in Mean-Time-Between-Failures (MTBF). The TCS shall have a specified θ_0 (θ_0 as defined in MIL-STD-781) MTBF of 290 hours and a minimum acceptable θ_1 MTBF of 193 hours when tested and accepted in accordance with the Quality Assurance Provisions, Section 4, of this specification.

3.3.3.4 Time totalizing meters. The CPS and TCA WRA's shall contain time totalizing meters in accordance with MIL-M-7793.

3.3.3.5 Parts reliability program. The contractor shall define, in his Reliability Program Plan, a parts reliability program to assure part reliability throughout the program. To the extent possible, the contractor shall use his standard policies, practices and operating procedures. The parts reliability program shall include:

a. Provide receiving test and inspection to assure uniform part quality as established and identified in the Program Plan.

b. Establish in-house and vendor surveillance over the application of parts, materials and processes during fabrication and assembly to ensure sources of degradation and variability are isolated and controlled.

c. Maintain visibility over failed parts analysis to ensure proper corrective action has been implemented.

d. Determine and evaluate program areas related to parts, materials, and processes, and initiate corrective action.

e. Review and analyze proposed changes in parts, materials, processes resulting from test, manufacturing, and field experience.

3.3.3.6 Electronic/electrical/electro-mechanical parts screening. The contractor shall provide for a 100 percent screening of all Electronic/Electrical/Electro-mechanical (EEE) parts which are not standard high reliability parts such as Established Reliability (ER); MIL-S-19500 JAN TX (Joint Army Navy, Testing Extra) semiconductor devices; MIL-M-38510 (Class B) microelectronics for EEE parts. Screening includes visual inspections, preconditioning and burn-in tests used to clarify discrepant parts. The contrac-

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tor shall screen all non-standard parts to the requirements of the closest standard high reliability parts. All integrated and hybrid circuits shall be screened in accordance with MIL-STD-883 Class B requirements.

3.3.3.7 Failure data. The contractor shall implement a closed-loop failure reporting analysis and corrective action system for reporting, analyzing, and accomplishing corrective action on equipment failure in accordance with MIL-STD-785A (see 6.2.2).

3.3.4 Maintainability. The Maintainability ($\bar{M}$) characteristics of the TCS system will be assured by a maintainability program using MIL-STD-470 as a guide. The program shall include maintenance concept, M prediction, and analysis. Actual maintainability shall be proven by comparing similarities to existing equipment.

3.3.4.1 Maintenance, ($\bar{M}$).

3.3.4.1.1 Maintenance concept. The maintenance concept for this equipment shall be divided into three categories; i.e., organization maintenance, intermediate maintenance, and depot maintenance. Organizational maintenance is defined as that maintenance performed on equipment while installed in the aircraft, including maintenance performed during periodic inspections. This maintenance shall be limited to functional or operational checks to determine quality of performance and operational readiness. Periodic checks are also required to ensure, to the maximum possible extent, that the equipment will operate throughout the next periodic level. Intermediate maintenance is defined as the maintenance required to isolate and repair faulty modules and subassemblies, adjustment, and alignment of components and equipments. Depot maintenance is the maintenance required to completely disassemble and perform detailed analysis and repair of equipment subassemblies and components.

3.3.4.1.2 Maintenance consideration. Maintenance consideration shall provide for the location of test points and the identification of procedures that will permit positive verification of equipment performance. These maintenance considerations and procedures shall provide for a quality of repair that will restore the equipment to an overall quality equal to or better than that required by the reliability and quality control requirements of this specification.

3.3.4.2 Repair cycle.

3.3.4.2.1 Repair time. Performance checks of equipment installed in the air vehicle shall not require more than 0.25 hour with two men, one a skilled mechanic (-5 level mechanic) and one an apprentice mechanic (-3 mechanic), using authorized flight line Peculiar Ground Support Equipment (PGSE) and the appropriate technical order. Shop maintenance following removal from aircraft at intermediate level, including necessary inspection and any subsequent repair required, shall not exceed 6.0 hours ($\bar{M}$ max ct) at 90th percentile using one skilled mechanic (-5 level mechanic). The equipment shall be repairable at field level by replacement of subassemblies, modules, or parts. Depot level maintenance on subassemblies or modules not repairable at field level shall not exceed 8 hours ($\bar{M}$ max ct), utilizing a technician, authorized depot PGSE, and technical information. Application of the foregoing numbers

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assumes the availability of adequate spare parts and facilities. The TCS system shall be designed to ensure that the Maintenance Manhour per Operating Hour (MMH/OH) will be 0.22 corrective and 0.01 preventive, and the Mean Time To Repair (MTTR) will be 2.0 hours at intermediate level.

3.3.4.2.2 Access and buttonup time. The MMH/OH and MTTR will not include time to gain access to aircraft components and to button up aircraft.

3.3.4.2.3 Movement time. The MMH/OH and MTTR will not include time to move personnel, spares or support equipment to and from maintenance stations.

3.3.4.2.4 Inactive time. The MMH/OH and MTTR will not include inactive maintenance time, e.g., drying of conformal coating on circuit boards.

3.3.4.2.5 Service and access. The equipment layout shall be constructed to provide ready access to all subassemblies, modules, and parts. Access to replaceable subassemblies and modules shall require minimal removal of any other components, except the outer case. Access to other subassemblies, modules, and parts shall require minimum time and effort of maintenance personnel. All parts requiring replacement at field level shall be constructed into or be a part of a replaceable subassembly or module. Test points shall be chosen to provide a straightforward, logical, step-by-step, troubleshooting sequence, as well as provide a single end-to-end performance check. Maintenance considerations shall provide for the identification of where and how to break loop or feedback circuits to allow for proper signal injection with test equipment, and shall provide a step-by-step troubleshooting, adjustment, or alignment of the circuit. Test points, where practical, shall be located so that all are accessible without moving or rotating the equipment on the bench. Adequate spacing shall be provided between test points, and between test points and other components, to eliminate the possibility of arcing, faulty readings or signals, or damage to equipment or components.

3.3.5 Environmental conditions. The TCS shall be designed to perform in the environmental conditions listed in the following paragraphs.

3.3.5.1 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25 °C ±10 °C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115/200 Vac 400 ±3% Hz, 4-wire, "Y"-connected

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3.3.5.2 Service conditions. The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2X equipment, except where aircraft installation location considerations exceed Class 2X levels.

3.3.5.3 Vibration. The TCS shall operate satisfactorily when subjected to the vibration profiles specified in Figures 13 through 18.

3.3.5.3.1 Vibration operating.

3.3.5.3.1.1 Random performance. The equipment shall meet the performance requirements specified in 3.3.1 while being subjected to the random vibration curve shown in Figure 13. The duration of the test shall be 10 minutes in each of three orthogonal axes, a longer time is permitted if required to verify system performance.

3.3.5.3.1.2 Sinusoidal endurance. The equipment shall meet the performance requirements specified in 3.3.1 after being subjected to the environment defined by the sinusoidal vibration curve shown in Figure 14. The equipment shall be operated during the test although to no specified tolerance. The duration of the test shall be 30 minutes in each of three orthogonal axes, one sweep from 10 to 400 to 10 Hz.

3.3.5.3.1.3 Random endurance. This test consists of superimposing three narrow bands of high level random on a low level broadband random spectrum. The specified narrow bands comprise a set of 36 contiguous frequency bands, each bandwidth being approximately 1/6 octave of the center frequency over the range of 20 to 2000 Hz. The center frequencies are specified in Table V. The duration of the test shall be one minute for each of the 36 steps specified in Table VI; a total of 36 minutes along each of three orthogonal axes. The equipment shall be on and operating throughout the test, although to no specified tolerance. After the test, the equipment shall meet the performance requirements of 3.3.1.

3.3.5.3.2 Vibration, nonoperating, simulated gunfire. The simulated gunfire test is based on a test method described in the Shock and Vibration Bulletin 41, Part 4, Page 37. The test environment simulates the complex wave periodic vibration impulses associated with rapid fire machine guns firing at nominal rates of 4000 and 6000 rounds per minute. The frequency content of these complex pulses is shown by the line spectra of Figures 17 and 18. The equipment shall operate satisfactorily after exposure to one minute of gunfire vibration at each of the 20 pulse rates given in Table VII.

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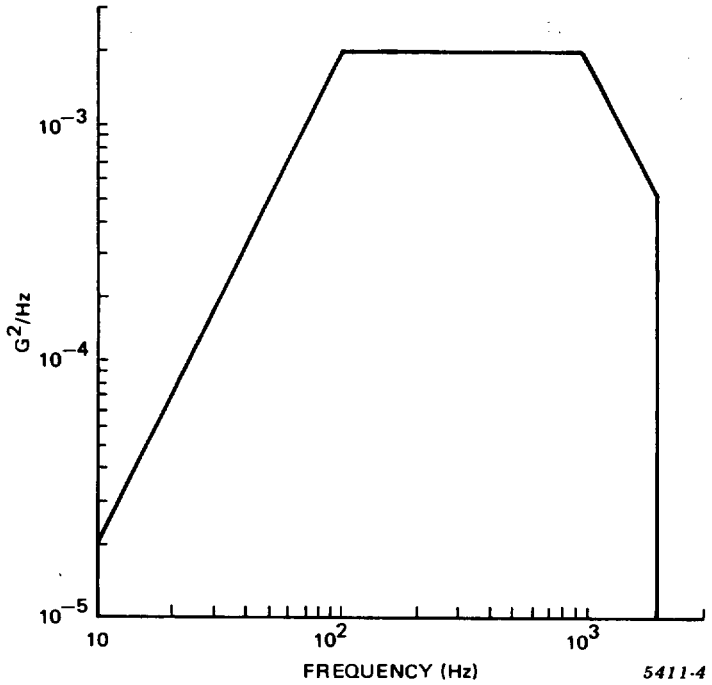


FIGURE 13. Random vibration performance test curve.

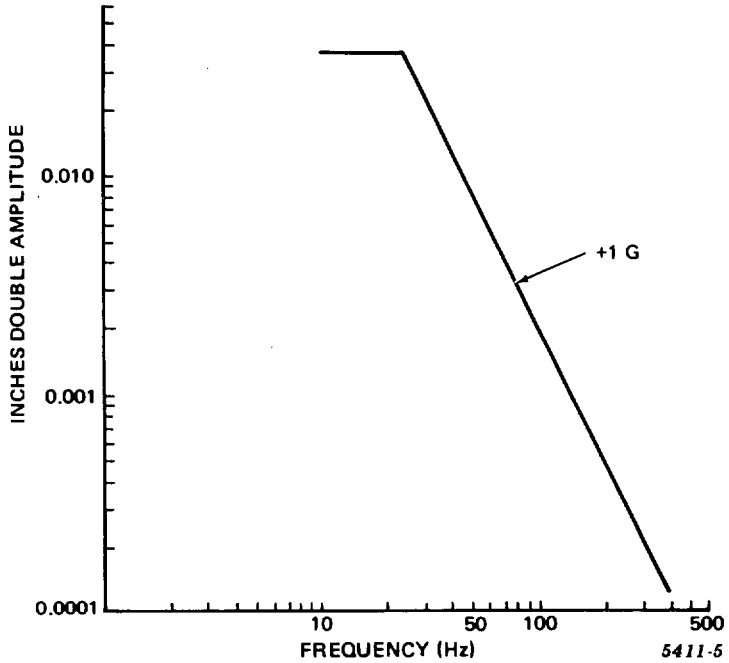


FIGURE 14. Endurance test curve.

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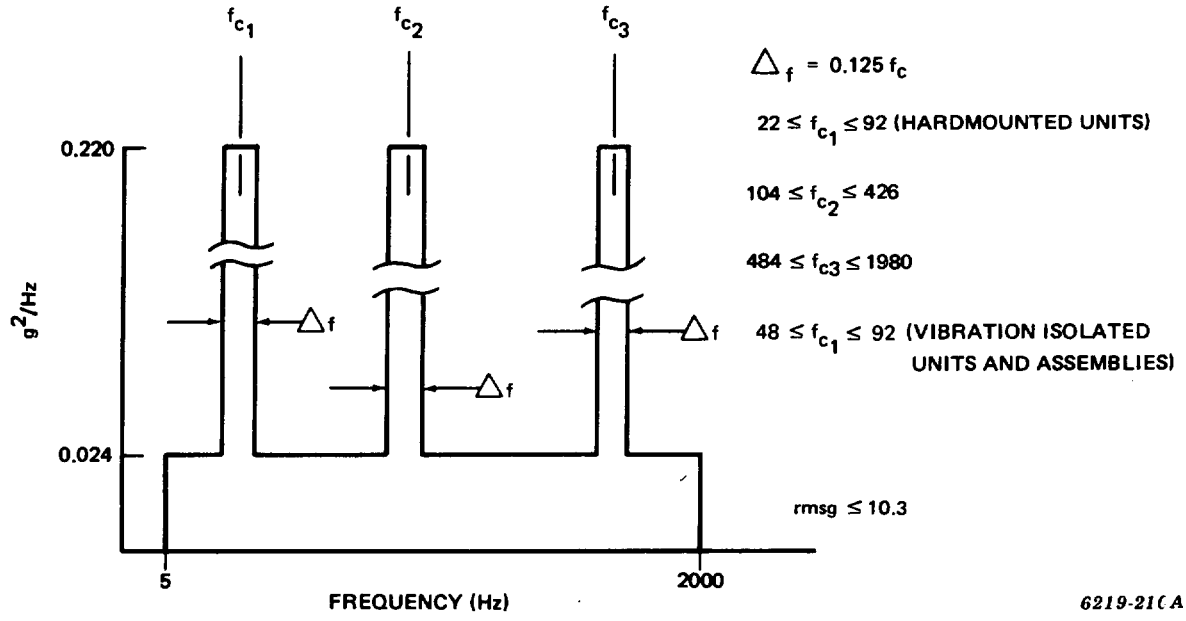


FIGURE 15. Acceleration spectral density, control-power supply (CPS).

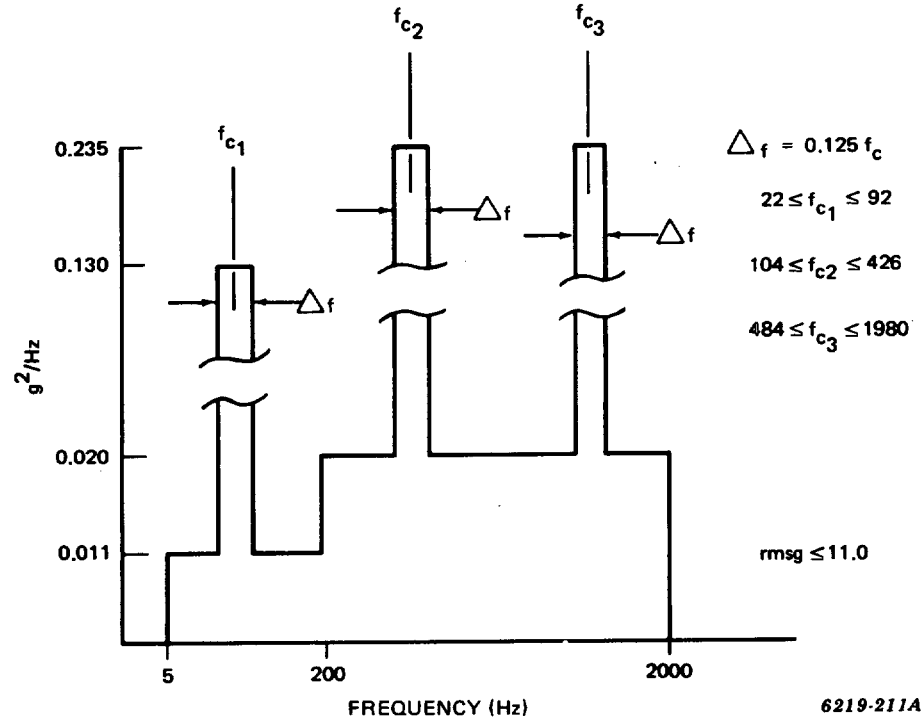


FIGURE 16. Acceleration spectral density, television camera assembly (TCA).

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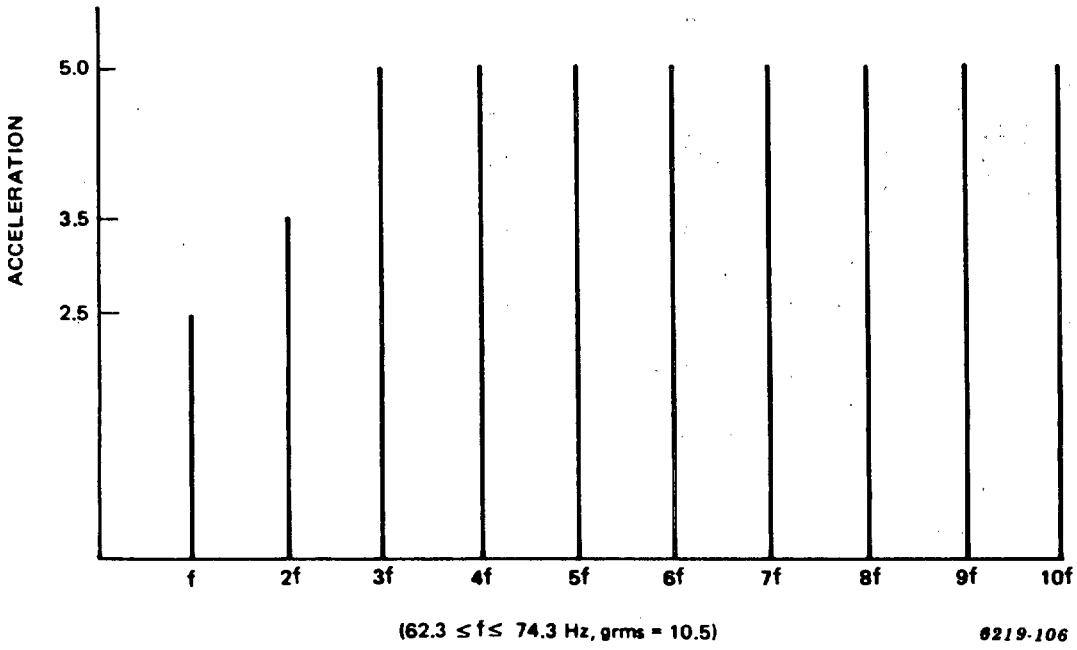


FIGURE 17. Gunfire vibration spectrum 4000 rpm nominal rate.

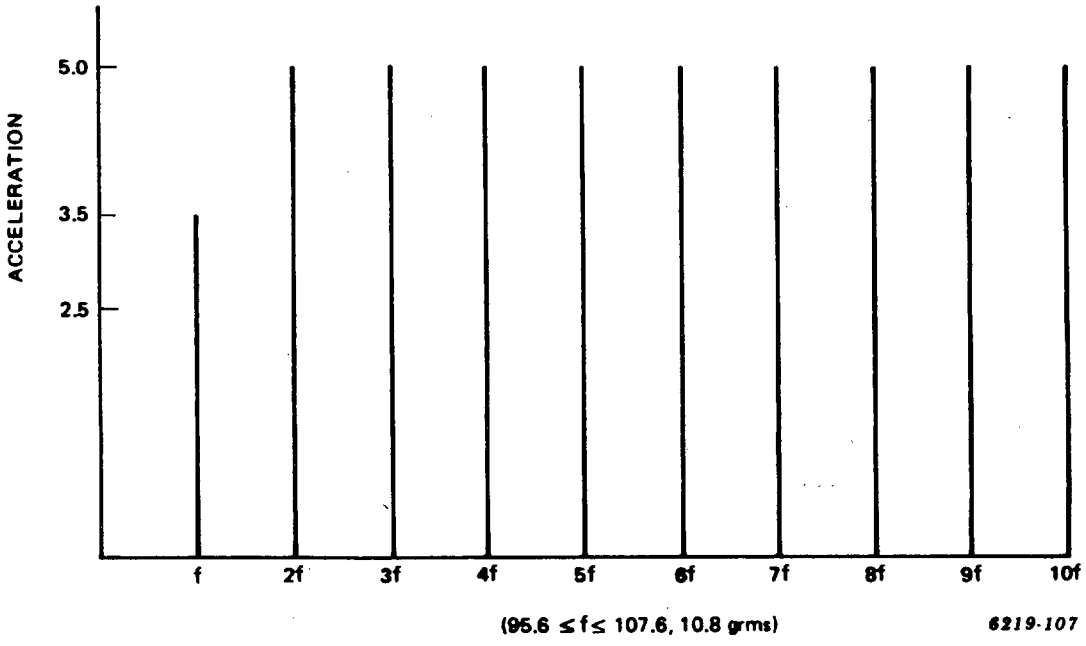


FIGURE 18. Gunfire vibration spectrum 6000 rpm nominal rate.

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TABLE V. Endurance random vibration, narrow band frequencies.

Frequency Band No.	Lower Frequency	Center Frequency	Upper Frequency
1	21	22	24
2	24	25	27
3	27	29	31
4	31	33	35
5	35	37	40
6	40	43	45
7	45	48	52
8	53	56	59
9	59	63	67
10	67	71	76
11	76	81	86
12	86	92	98
13	98	104	111
14	111	119	126
15	126	135	144
16	144	153	163
17	163	174	186
18	186	198	211
19	211	225	240
20	240	256	273
21	273	290	310
22	310	330	352
23	352	375	400
24	400	426	455
25	455	484	517
26	517	550	587
27	587	626	667
28	667	711	759
29	759	808	862
30	862	919	980
31	980	1044	1113
32	1113	1187	1265
33	1265	1349	1438
34	1438	1533	1634
35	1634	1742	1857
36	1857	1980	2110

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TABLE VI. Endurance random vibration, schedule of stepped frequency combinations.

Time Step No. (1 minute Vib./Step)	Frequency Band No. (Table V)			
	1/ f_{c_1}	2/ f_{c_1}	f_{c_2}	f_{c_3}
1	12	12	24	36
2	11	11	23	35
3	10	10	22	34
4	9	9	22	34
5	8	8	23	35
6	7	--	24	36
7	7	--	24	36
8	8	8	23	35
9	9	9	22	34
10	10	10	21	33
11	11	11	20	32
12	12	12	19	31
13	12	12	19	30
14	11	11	20	29
15	10	10	21	28
16	9	9	21	27
17	8	8	20	26
18	7	--	19	25
19	6	--	18	25
20	5	--	17	26
21	4	--	16	27
22	4	--	15	28
23	5	--	14	29
24	6	--	13	30
25	6	--	13	31
26	5	--	14	32
27	4	--	15	33
28	3	--	16	33
29	2	--	17	32
30	--	--	18	31
31	--	--	18	30
32	2	--	17	29
33	3	--	16	28
34	3	--	15	27
35	2	--	14	26
36	--	--	13	25

Notes: 1/ Band numbers applicable to the TCA and hardmounted units of the CPS. The dash (--) denotes f_{c_1} shall not be applied.

2/ Band numbers applicable to vibration isolated units of the CPS.

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TABLE VII. Pulse rates for gunfire vibration simulation.

Time Step No. (1 min Vib./Step)	Low Firing Rate (4000 rpm) Frequency	High Firing Rate (6000 rpm) Frequency
1	62.3 ±0.1	95.6 ±0.1
2	63.5 ±0.1	96.9 ±0.1
3	64.8 ±0.1	87.2 ±0.1
4	66.1 ±0.1	99.5 ±0.1
5	67.4 ±0.1	100.8 ±0.1
6	68.7 ±0.1	102.1 ±0.1
7	70.0 ±0.1	103.4 ±0.1
8	71.4 ±0.1	104.8 ±0.1
9	72.8 ±0.1	106.2 ±0.1
10	74.3 ±0.1	107.6 ±0.1

3.3.5.4 Temperature. The equipment shall meet the requirements of MIL-E-5400, Class 2X, except that the maximum nonoperating temperature shall be limited to 90 °C.

3.3.5.5 Altitude. Both weapon replaceable assemblies shall be pressurized to approximately one atmosphere and shall meet requirements of 3.3.1 throughout the entire flight envelope of the F-14 aircraft.

3.3.5.6 Shock loads.

3.3.5.6.1 Shock, operating. The equipment shall meet the performance requirements of 3.3.1 after exposure to three 15 g, 11 ms half sine shock pulses in each direction along each of three orthogonal axes for a total of 18 shocks. The equipment shall be on and operating during the test, although to no specified tolerance.

3.3.5.6.2 Shock, nonoperating. The equipment shall operate satisfactorily after being subjected to two 25 g, 3 ms half sine shock pulses in each direction along each of three orthogonal axes for a total of 12 shocks. In addition, the units shall withstand a half sine crash safety shock of 30 g, 11 ms. The units shall be exposed to two shocks in each direction along each of three orthogonal axes for a total of 12 shocks. Bending and distortion are permitted but no parts shall become adrift.

3.3.5.7 Humidity. The TCS system equipment shall be designed to meet the humidity requirements of MIL-E-5400.

3.3.5.8 Maneuver loads.

3.3.5.8.1 Normal loads. The TCS equipment shall be designed to perform in accordance with 3.3.1 in a simulated mounting, while being subjected to the following loads:

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a.	Vertical Down	4.0 g
b.	Vertical Up	1.5 g
c.	Horizontal	±3.0 g
d.	Forward	4.0 g
e.	Aft	4.0 g

3.3.5.8.2 Limit loads. The TCS equipment shall be designed to meet the performance requirements of this specification after application of limit load factors. The equipment shall operate during the application of the load factors. No physical distortion or permanent set of the equipment is permitted after application of limit load factors.

a.	Vertical Down	9.6 g
b.	Vertical Up	5.2 g
c.	Horizontal	±3.7 g
d.	Forward	4.0 g
e.	Aft	5.0 g

3.3.5.8.3 Ultimate loads. The equipment or equivalent dummy loads shall remain in place, without failure of the attaching points, when subjected to ultimate loads. Distortion and permanent set are permitted and equipment operation is not required during or after application of ultimate loads.

a.	Vertical Down	15.6 g
b.	Vertical Up	6.15 g
c.	Horizontal	±9.15 g
d.	Forward	4.0 g
e.	Aft	5.0 g

3.3.6 Transportability. The TCS system shall be suitable for air cargo transportation.

3.4 Design and construction. The equipment shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.4.1 Parts and materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

b. Other parts and materials requirements shall be in accordance with MIL-E-5400, Class 2X, Category III.

c. Nonrepairable subassemblies shall be used in accordance with MIL-E-5400.

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d. When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

e. Conformal coatings shall be approved types and meet the requirements of MIL-I-46058 and MIL-E-5400. Use of the insulating compound known as "Norcote," in accordance with WS-1308, shall be permitted.

3.4.1.1 Modules. The electronic portions of the equipment shall be functionally modularized in accordance with MIL-E-5400.

3.4.2 Electromagnetic Interference (EMI) control. The contractor's equipment for the TCS system shall incorporate adequate Electromagnetic Interference (EMI) design in such a manner that the equipment complies with the EMI requirements of MIL-STD-461, for Class A-1 equipment, as follows: CE03, CE04, CS01, CS02, CS06, RE02, RS02, and RS03. Radiation susceptibility tests are applicable to that portion of the video sensor assembly exposed to the air stream. AFSC Design Handbook DH 1-4 shall be used as a design guide to control the EMI characteristics (emission and susceptibility) of the TCS equipments.

3.4.3 Nomenclature, nameplates and identification marking. Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with MIL-N-18307.

3.4.4 Interchangeability. The equipment shall meet the interchangeability requirements of MIL-E-5400. All Weapon Replaceable Assemblies (WRA's) with the same part or identification number shall be interchangeable as to form, fit, and function.

3.4.5 Safety.

3.4.5.1 Flight safety. The TCS system equipment shall meet the explosion requirements of MIL-E-5400.

3.4.5.2 Ground safety. The TCS system equipment shall meet the explosion requirements of MIL-E-5400.

3.4.5.3 Personnel safety. The TCS shall meet the personnel safety requirements of MIL-E-5400.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests. Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

a. First Article Test (see 4.4.2.4)
b. Acceptance Tests (see 4.4)

4.2 Scope of tests. Production tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of MIL-T-5422.

4.3 Performance requirements verification. The performance requirement is shown in Table VIII.

4.4 Acceptance tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the surveillance of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall generate test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product.

4.4.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of materials, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of Product
b. Operational Test

4.4.1.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

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TABLE VIII. Performance requirement verification.

Paragraph	Requirement	Verified 1/	Inspection 2/	Analysis	Acceptance Test 3/
3.1.1	Major component list	X			
3.3.1	Detail performance requirement	-	-	-	-
3.3.1.1	Television Camera Assembly (TCA)	-	-	-	-
3.3.1.1.1	Function	X			
3.3.1.1.2	Form factor	X	X		
3.3.1.1.3	Weight	X	X		
3.3.1.1.4	Contents	-	-	-	-
3.3.1.1.5	Controls	-	-	-	-
3.3.1.1.6	Electrical connections	X	X		
3.3.1.1.7	Warmup time	X	X		
3.3.1.2	Control-Power Supply (CPS)	-	-	-	-
3.3.1.2.1	Form factor	X	X		
3.3.1.2.2	CPS performance	X			
3.3.1.3	Television Camera Set (TCS)	-	-	-	-
3.3.1.3.1	Tracking mode	X			X
3.3.1.3.2	Acquisition modes	X			X
3.3.1.3.2.1	Manual acquisition	X		X	X
3.3.1.3.2.2	Automatic acquisition	X		X	X
3.3.1.3.2.3	Automatic acquisition with search	X			X
3.3.1.3.2.4	Search patterns	X			X
3.3.1.3.2.5	Semiautomatic acquisition	X			X
3.3.1.4	Instantaneous FOV	X			X
3.3.1.5	Cone of vision	X			X
3.3.1.6.1	Display centerline	X		X	
3.3.1.6.2	Track window alignment	X			X
3.3.1.6.3	Bench alignment accuracy	X			X
3.3.1.6.4	Bench boresight adjustments	X			X
3.3.1.7	Lines of sight	X			X
3.3.1.8	Sun sensor	X		X	X
3.3.1.8.1	Sun shutter triggering levels	X			X
3.3.1.8.2	Sun sensor cone of view	X			
3.3.1.8.3	Sun sensor shutter speed	X			
3.3.1.9	Gimbal slewing dynamics	X			X
3.3.1.10	Automatic video tracking target dynamics	X			
3.3.1.11	Base Motion Isolation	X			
3.3.1.12	Optical video tracker	X			X

1/ Verified by test and analysis during preproduction program
2/ Includes subsystem calibration
3/ Northrop Acceptance Test Procedure No. 336700 is used

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TABLE VIII. Performance requirement verification. - Continued

Paragraph	Requirement	Verified 1/	Inspection 2/	Analysis	Acceptance Test 3/
3.3.1.13	Static limiting resolution	X			X
3.3.1.14	Dynamic limiting resolution	X			
3.3.1.15	Square wave response	X			X
3.3.1.16	Gray scale rendition	X			X
3.3.1.17	Automatic light range	X			X
3.3.1.18	Scan failure protection	X		X	
3.3.1.19	TCS geometric distortion	X	X		
3.3.1.20	Television format	X			X
3.3.1.21	Display symbology				X
3.3.1.22	Angular accuracy				X
3.3.1.24	Radome refraction compensation				X
3.3.1.25	TCS reset				X
3.3.2	TCS/F-14 interface	-	-	-	-
3.3.2.1	Video outputs	X			X
3.3.2.2	Horizontal sync signal	X			X
3.3.2.3	Vertical sync signal	X			X
3.3.2.4	TCS onboard signal	X			X
3.3.2.5	TCS Temperature warning signal	X	X		
3.3.2.6	TCS elevation gimbal	X			X
3.3.2.7	TCS optical position to radar	X			X
3.3.2.8	TCS optical position to AN/AWG-9 computer	X			X
3.3.2.9	TCS gimbal angle from AN/AWG-9 computer	X			X
3.3.2.10	Computer return #1	X	X		
3.3.2.11	SPATS (Search Pattern Size)	X			X
3.3.2.12	+10 Vdc analog reference	X			X
3.3.2.13	TCS track valid	X			X
3.3.2.14	TCS azimuth gimbal	X			X
3.3.2.15	Computer return #2	X	X		
3.3.2.16	TCS power interlock signal	X			X
3.3.2.17	TCS inputs from AN/AWG-9 and F-14	-	-	-	-
3.3.2.17.1	Hand control X	X			X
3.3.2.17.2	Hand control Y	X			X
3.3.2.17.3	Half action	X			X
3.3.2.17.4	Full action	X			X
3.3.2.17.5	TCS power on select	X			X

1/ Verified by test and analysis during preproduction program
2/ Includes subsystem calibration
3/ Northrop Acceptance Test Procedure No. 336700 is used

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TABLE VIII. Performance requirement verification. - Continued

Paragraph	Requirement	Verified 1/	Inspection 2/	Analysis	Acceptance Test 3/
3.3.2.17.6	AUTO-ACQ	X			X
3.3.2.17.7	AUTO-SEARCH	X			X
3.3.2.17.8	RDR/TCS trim azimuth	X			X
3.3.2.17.9	RDR/TCS trim elevation	X			X
3.3.2.17.10	TCS standby select	X			X
3.3.2.17.11	TCS slave select	X			X
3.3.2.17.12	Radar tracking	X			X
3.3.2.17.13	104 Vrms, 400 Hz Ref.	X			X
3.3.2.17.14	Rear video select	X			X
3.3.2.17.15	Front video select	X			X
3.3.2.17.16	SLAC (Slave to Computer Command)	X			X
3.3.2.18	TCS analog inputs from Computer Signal Data Converters (CSDC)	-	-	-	-
3.3.2.18.1	Antenna position	X			X
3.3.2.19	TCS coolant input from F-14	X	X		
3.3.2.20	Input electrical power	-	-	-	-
3.3.2.20.1	Operating power	X			
3.3.2.20.2	Lighting power	-	-	-	-
3.3.2.20.3	Degraded performance	X			
3.3.2.20.4	Standby power	X	X		
3.3.2.21	TCS temperature warning signal	X	X		
3.3.2.22.1	Cables and connectors	X	X		
3.3.2.22.2	Interconnecting cabling	X	X		
3.3.3	Reliability	X		X	
3.3.3.1	Operational stability	X			
3.3.3.2	Operating life			X	
3.3.3.3	Reliability in Mean-Time-Between-Failure (MTBF)	X		X	
3.3.3.4	Time totalizing meter	X	X		
3.3.3.5	Parts reliability program			X	
3.3.3.6	Electronic/electrical/ electro-mechanical parts screening	X	X		
3.3.3.7	Failure data	-	-	-	-
3.3.4	Maintainability			X	
3.3.4.1	Maintenance, ($\bar{M}$)			X	

1/ Verified by test and analysis during preproduction program
2/ Includes subsystem calibration
3/ Northrop Acceptance Test Procedure No. 336700 is used

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TABLE VIII. Performance requirement verification. - Continued

Paragraph	Requirement	Verified 1/	Inspection 2/	Analysis	Acceptance Test 3/
3.3.4.1.1	Maintenance concept			X	
3.3.4.1.2	Maintenance consideration	X			
3.3.4.2	Repair cycle	-	-	-	-
3.3.4.2.1	Repair time	X			
3.3.4.2.2	Access and buttonup time	-	-	-	-
3.3.4.2.3	Movement time	-	-	-	-
3.3.4.2.4	Inactive time	-	-	-	-
3.3.4.2.5	Service and access	X			
3.3.5	Environmental conditions	X			
3.3.5.1	Standard conditions	X	X		
3.3.5.2	Service conditions	X			
3.3.5.3	Vibration	X		X	
3.3.5.3.1	Vibration operating	-	-	-	-
3.3.5.3.1.1	Random performance	X		X	
3.3.5.3.1.2	Sinusoidal endurance	X		X	
3.3.5.3.1.3	Random vibration endurance	X		X	
3.3.5.3.2	Vibration, nonoperating, simulated gunfire	X		X	
3.3.5.4	Temperature	X		X	
3.3.5.5	Altitude	X		X	
3.3.5.6	Shock loads	X		X	
3.3.5.6.1	Shock, operating	X		X	
3.3.5.6.2	Shock, nonoperating	X		X	
3.3.5.7	Humidity	X		X	
3.3.5.8	Maneuver loads	-	-	-	-
3.3.5.8.1	Normal loads	X		X	
3.3.5.8.2	Limit loads	X		X	
3.3.5.8.3	Ultimate loads	X		X	

1/ Verified by test and analysis during preproduction program
2/ Includes subsystem calibration
3/ Northrop Acceptance Test Procedure No. 336700 is used

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4.4.1.2 Operational test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.2 Reliability assurance tests. The Reliability Assurance Tests shall consist of the following:

a. In Process Screening
b. Preconditioning (Burn-In)
c. Reliability Production Acceptance Test
d. First Article Test
e. Special Test

4.4.2.1 In-process screens. The contractor shall perform lower level assembly screens (including inspection, test or burn-in) consistent with good manufacturing practices and in the most cost effective manner in order to permit detecting the maximum number of discrepancies prior to system tests. The establishment of production screening points is dependent on equipment design, quality level of parts, assembly flow and critical fabrication points. Screening shall include nonfunctional vibration of all PWB's. The vibration spectrum shall be per Figure 19 for 10 minutes in the axis perpendicular to the printed wiring boards.

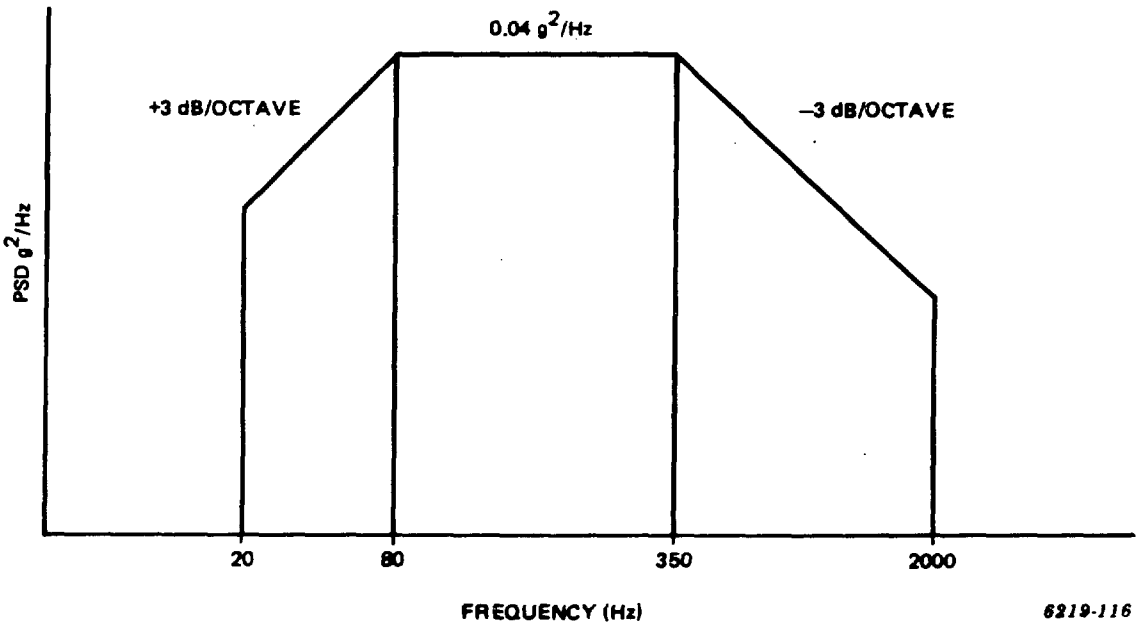


FIGURE 19. Random vibration spectrum.

4.4.2.2 Preconditioning (burn-in). Each system shall be exposed to the following thermal and vibration screening.

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4.4.2.2.1 Thermal screening. Each system shall be subjected to a minimum of 10 temperature cycles from -54°C to +71°C to -54°C at a minimum average rate of change of 5°C/minute (the contractor may adjust the temperature values provided the range from minimum to maximum is not less than 125°C). Dwell time at either end of temperature extreme shall be at option of contractor. Each system shall be operating and subjected to voltage cycling during periods of increasing temperature and nonoperating during decreasing temperature. The system shall be functionally tested once during operation and continuously monitored. Normal system cooling provisions shall be provided. The last test cycle shall be failure free.

4.4.2.2.2 Vibration. Each system shall be subjected to 10 minutes of random vibration. The equipment shall be hard mounted to a vibration table capable of reproducing random vibration having the power spectral density characteristics shown in Figure 19. A pseudo-random or synthetic vibration shaker capable of reproducing this power spectral density function will be an acceptable substitute for a true random vibration fixture. The equipment under tests should be oriented on the fixture such that the axis of vibration is perpendicular to the carrier boards.

4.4.2.3 Production reliability acceptance test. Following completion of preconditioning (burn-in), each system shall be subjected to a production reliability acceptance test. The test environment shall be the same as that defined in 4.4.2.2.1 and 4.4.2.2.2. Each system shall accumulate 50 hours of operating time and shall be failure free in the last two cycles. The Production Reliability Acceptance Test Plan derived from Test Plan IIC, as defined in Figure C-11 of MIL-STD-781 shall be used to evaluate continuing MTBF acceptability.

4.4.2.3.1 Failure criteria. The failure criteria for the Production Reliability Acceptance Test shall be in accordance with Paragraph 6.0 inclusive of drawing 336241.

4.4.2.4 First article test. Two production systems randomly selected from the initial ten production units shall be subject to 600 operating hours (cumulative total) of the Reliability Qualification (Demonstration) Test environment (see drawing 336256) to assure the continued integrity of production hardware. No one system shall accumulate more than 350 hours. The test will be successfully completed if no failures requiring design modification are experienced, and the last 20 cycles are failure free. In the event that any failure requiring design modification occurs, the contractor shall implement the corrective action on the test hardware and all other production systems, and demonstrate the effectiveness of the corrective action by continuing the test for 200 hours without failure of the design modification. The test shall be extended beyond 600 hours if required to fulfill the 200-hour verification requirement.

4.4.2.4.1 Test details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be

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generated and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests (see 6.2.2).

4.4.2.5 Special tests. These shall be conducted when any class I design or material change affects the performance of the equipment.

4.4.2.5.1 Scope of test. Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved shall be used where applicable. When previously approved test procedures do not exist, the contractor shall generate test procedures and obtain procuring activity approval prior to conducting the tests.

4.4.2.6 Equipment failure. Should a failure occur during the above reliability assurance tests, the following action shall be taken:

a. Determine the cause of failure (see 6.2.2).

b. Determine if the failure is an isolated case or design defect.

c. Propose corrective action intended to reduce the possibility of the same failure occurring in future tests.

d. Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test procedures. The procedures used for conducting acceptance tests shall be prepared by the contractor. The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.6 Spare equipment provisioning (WRA/SRA) acceptance. Unless otherwise specified by the contract, it is intended that the items covered by this specification shall be accepted at the system level of assembly, to ensure equipment compatibility and to ensure that the Maintainability requirements of 3.3.4 and the Interchangeability requirements of 3.4.4 are met. Acceptance of spare WRA's and SRA's may be performed at lower levels of assembly than the system level, provided that the interchangeability of spare items procured under the contract is adequately demonstrated by the contractor.

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5. PACKAGING

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended use. The TCS is intended for installation on the F-14 aircraft to provide enhanced visual target identification and tracking.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. Selection of applicable levels of packaging and packing (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph	Data Requirement	Applicable DID
3.3.3.2	Provisioning Parts List	DI-R-7079
3.3.3.7	Report, Failure Summary and Analysis	DI-R-7041
4.4	Acceptance Test Reports	UDI-T-23967
4.4.2.4	First Article Test Procedure	UDI-T-21347
4.4.2.4.1	First Article Test Procedure	UDI-T-21347
4.4.2.6	Failure Report	UDI-R-21141A
4.5	Test Procedures	UDI-T-21347

6.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence should apply:

a. Contract. The contract should have precedence over any specification.

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b. This Specification. This specification should have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, should be specifically approved in writing by the procuring activity.

c. Referenced Specification. Any referenced specification should have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications should apply to the extent specified.

6.4 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which should be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval should be submitted to the procuring activity for consideration. Each request should be accompanied by complete supporting information.

6.5 Type designations. The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.4.3. The correct type number should be used on nameplates, shipping records, and instruction books, as applicable.

6.6 First article. When a first article is required, the item will be tested and should be a sample selected from the first ten production items. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, and test approval of the document's first article.

6.7 Glossary. A glossary of terms used herein is listed in Appendix C.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity
Navy-AS
(Project 5821-N160)

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APPENDIX A
CONTROL SWITCH MODE CHART.

Control Switches				Logic Discrete		Function
Slave	Acquisition	Action Switch	FOV	RTG	SLAC	
TCS	AS	ZA	NARROW	$\overline{\text{RTG}}$	SLAC	TSC slaves to computer and commences automatic scan. Locks automatically on first target seen in NFOV.
RADAR	--	ZA	--	--	--	Radar slaves to TCS if TCS is tracking. If not tracking, TCS is caged.
IND	AA	ZA	--	--	--	TCS is caged.
IND	AA	HA	--	--	--	Semiautomatic acquisition mode. TCS LOS controlled by hand control. FA initiates auto-acquisition of target in FOV.
IND	AS	ZA	--	--	--	TCS is caged.
IND	AS	HA	--	--	--	Semiautomatic acquisition mode. TCS LOS controlled by hand control. FA initiates auto-acquisition of target in FOV.
TCS	MANUAL	ZA	--	RTG	$\overline{\text{SLAC}}$	TCS slaves to radar LOS. NFO must manually acquire target lock-on by going to HA, bringing the target into the tracking window, and going to FA to initiate the tracker.
TCS	MANUAL	ZA	--	$\overline{\text{RTG}}$	SLAC	TCS slaves to computer. NFO must manually acquire target lock-on, as above.
TCS	AA	ZA	--	RTG	SLAC	TCS slaves to computer and automatically locks on target in FOV. In TCS, SLAC has priority over RTG.
TCS	AA	ZA	WIDE	RTG	$\overline{\text{SLAC}}$	TCS slaves to radar LOS and automatically locks on target in WFOV.
TCS	AA	ZA	NARROW	RTG	$\overline{\text{SLAC}}$	TCS slaves to radar LOS and automatically locks on target in NFOV.

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APPENDIX A
CONTROL SWITCH MODE CHART. - Continued

Control Switches				Logic Discrete		Function
Slave	Acquisition	Action Switch	FOV	RTG	SLAC	
TCS	AA	HA	--	--	--	Semiautomatic acquisition mode. TCS LOS controlled by hand control. FA initiates auto-acquisition of target in FOV.
TCS	AS	ZA	WIDE	RTG	$\overline{\text{SLAC}}$	TCS slaves to radar LOS and commences automatic scan. Locks automatically on first target seen in WFOV.
TCS	AS	ZA	NARROW	RTG	$\overline{\text{SLAC}}$	TCS slaves to radar LOS and commences automatic scan. Locks automatically on first target seen in NFOV.
TCS	AS	HA	--	--	--	Semiautomatic acquisition mode. TCS LOS controlled by hand control. FA initiates auto-acquisition of target in FOV.
TCS	AA	ZA	WIDE	$\overline{\text{RTG}}$	SLAC	TCS slaves to computer generated LOS and automatically locks on target in WFOV.
TCS	AA	ZA	NARROW	$\overline{\text{RTG}}$	SLAC	TCS slaves to computer generated LOS and automatically locks on target in NFOV.
TCS	AS	ZA	WIDE	$\overline{\text{RTG}}$	SLAC	TCS slaves to computer and commences automatic scan. Locks automatically on first target seen in WFOV.

NOTE: See Appendix C for a glossary of terms

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APPENDIX B
DIGITAL TYPE A SIGNAL CHARACTERISTICS.

20.1 Digital type A avionics receiver characteristics. The TCS circuitry shall incorporate a digital Type A avionics receiver and shall respond to signals generated by a digital Type A transmitter. The digital Type A receiver shall be capable of receiving digital signals having the following characteristics:

a.	Logical "one"	+5 to +14 Vdc
b.	Logical "zero"	-2 to +2 Vdc or open circuit
c.	Pulse Width	6 microseconds minimum in either a logical "one" or "zero" state. This minimum may be reduced to 3 microseconds provided the total of rise and fall times does not exceed 0.5 microsecond.
d.	Receiver load	1500 ohms ±30 percent resistive and 100 pf capacitance maximum to ground for single loads, but is individually specified for multiple loads.
e.	Requirements	Receiver shall not be dependent on the transmitter accepting current or providing a minimum source resistance.

20.2 Digital type A avionics/transmitter characteristics. Performance of the TCS digital Type A receiver assumes that the AN/AWG-9 radar/computer incorporates a digital Type A transmitter. The assumed digital Type A transmitter will be capable of providing digital signals having the following characteristics when terminated in a resistive load of 1500 ohms ±30 percent:

a.	Logical "one"	+6 to +13 Vdc
b.	Logical "zero"	-1 to +1 Vdc or open circuit
c.	Rise and fall times (measured between 10 and 90 percent points)	0.5 to 3 microseconds when resistance is in parallel with a 680 picofarad capacitance: 6 microseconds maximum when resistance is in parallel with 3000 picofarads maximum (includes load and wiring capacitance)
d.	Pulse width	Individually specified for each signal but shall be a minimum of 6 microseconds in either a logical "one" or "zero" state. 6 microseconds minimum can be reduced to 3 microseconds provided the total of rise and fall times does not exceed 0.5 microsecond.

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APPENDIX B
DIGITAL TYPE A SIGNAL CHARACTERISTICS. - Continued

e.	Short circuit protection	Any value of load including short circuit to ground shall not result in damage to the circuit components or affect any other channel.

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APPENDIX C
GLOSSARY.

Action Switch	-	Three-position switch on rate stick. It is used to enable rate control and track initiation
AA	-	Automatic Acquisition (Auto-Acquisition) to acquire and track a target without operator assistance
AS	-	Automatic Search (Auto-Search).
Break Lock	-	To terminate a track mode of operation
Caged	-	When the TCS is internally slaved to a fixed forward position
CPS	-	Control-Power Supply
CSDC	-	Computer Signal Data Converter
DD	-	Digital Display
DDD	-	Detailed Data Display
EMI	-	Electromagnetic Interference
FA	-	Full Action - To completely actuate the action switch to second detent
FOV	-	Field of View
GACH	-	Gimbal Angle Crosshair
Gimbal Limit	-	The mechanical limit of travel of the TCS Az and El gimbals
HA	-	Half Action - To close the action switch to its first detent
Horizontal Period	-	Time required for one TV horizontal sweep and retrace = 1/15,750 $\pm$0.25% sec
Horizontal Shading	-	Distortion in the video of a horizontal sweep
Ind	-	Independent
LOS	-	Line of Sight
MTBF	-	Mean Time Between Failures
NFO	-	Naval Flight Officer

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APPENDIX C
GLOSSARY. - Continued

NFOV	-	Narrow Field of View
PPLS	-	Program Parts Selection List
PWB	-	Printed Wiring Board
RACH	-	Radar LOS Position Crosshair
RDR	-	Radar
RTG	-	Radar Tracking discrete logic signal
SLAC	-	Slave to Computer Command discrete logic signal
SPATS	-	Search Pattern Size discrete logic signal
SRA	-	System Replaceable Assembly
TCA	-	Television Camera Assembly
TCS	-	Television Camera Set
TID	-	Tactical Information Display
TVL	-	Television Line
TVSU	-	Television Sight Unit
TW	-	Track Window
VDI	-	Vertical Display Indicator
Vertical Period	-	The time required for one TV vertical sweep and retrace = 1/60 sec $\pm$0.25%
WFOV	-	Wide Field of View
WRU	-	Weapon Replaceable Unit
ZA	-	Zero Action

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